Topic 5

Learning

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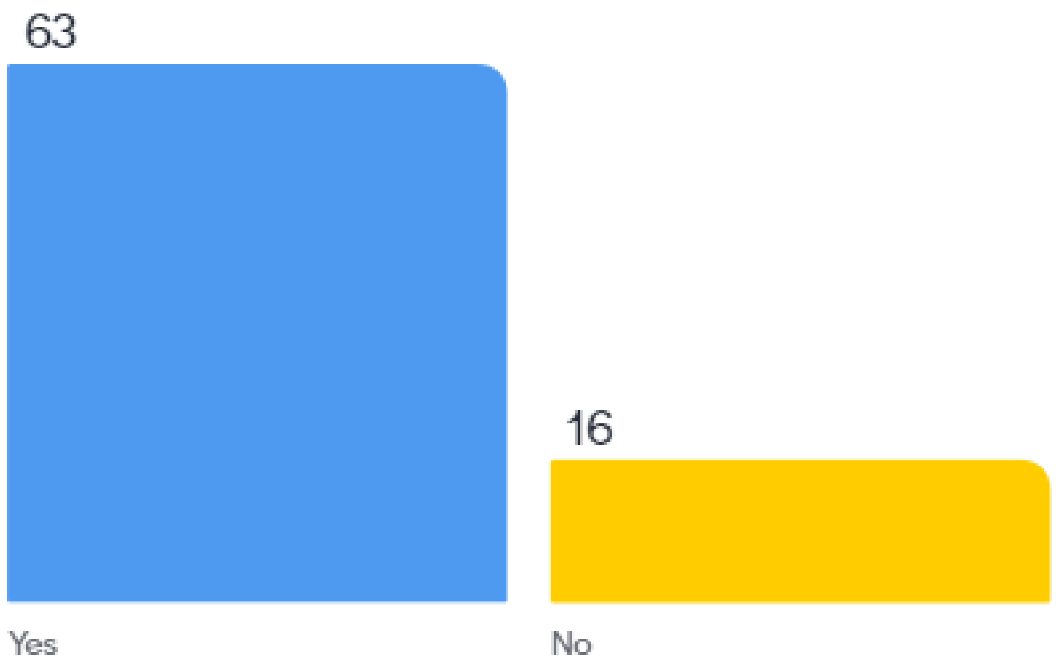
Do you agree with this statement:

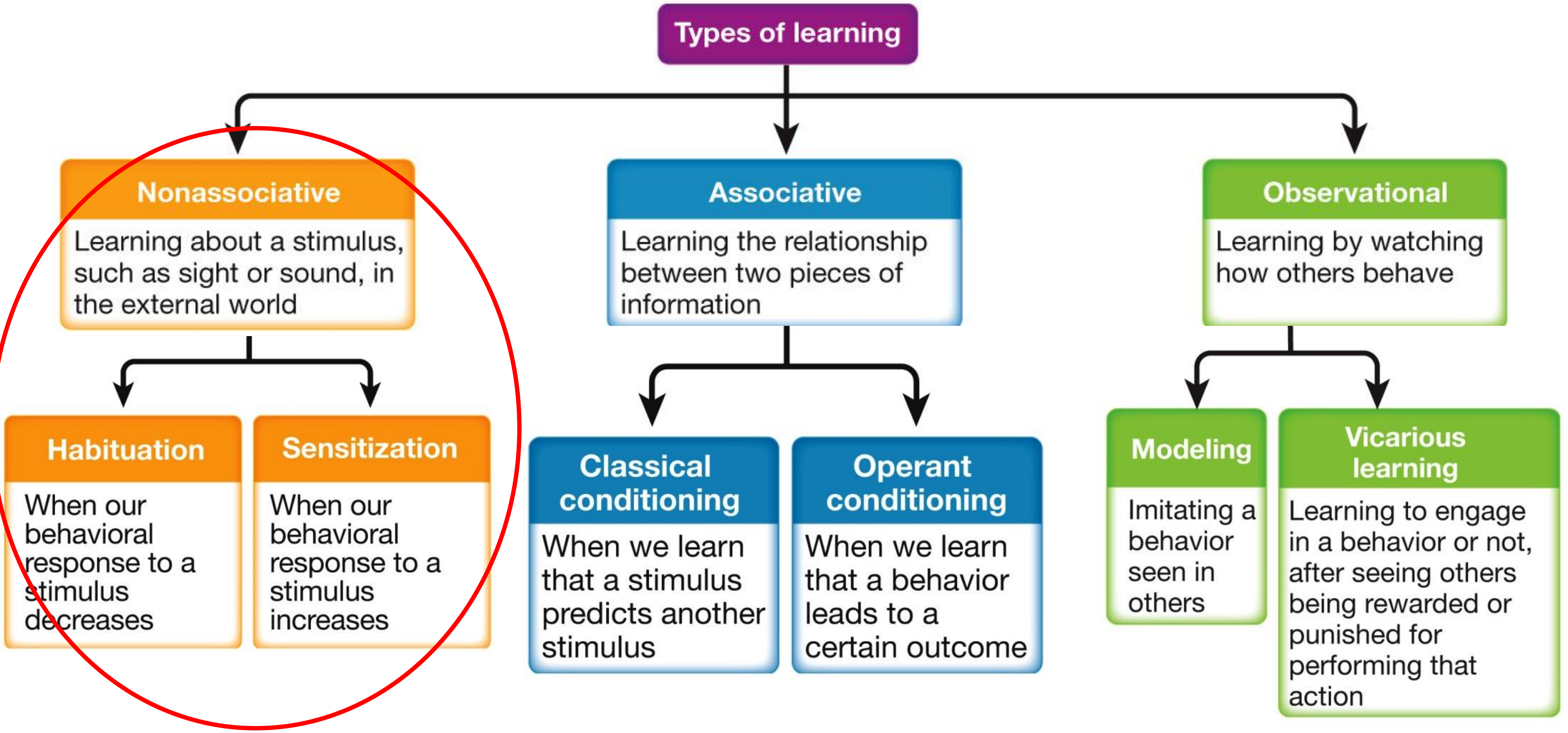
"Give me a dozen healthy infants, well formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select—doctor, lawyer, artist, merchant-chief, and yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations and race of his ancestors"



Overview

- Non-associative learning
- Associative learning: **classical** conditioning
- Associative learning: **operant** conditioning
- Observational learning

Overview


Types of learning
Nonassociative
Learning about a stimulus, such as sight or sound, in the external world
Associative
Learning the relationship between two pieces of information
Observational
Learning by watching how others behave
Habituation
When our behavioral response to a stimulus decreases
Sensitization
When our behavioral response to a stimulus increases
Classical conditioning
When we learn that a stimulus predicts another stimulus
Operant conditioning
When we learn that a behavior leads to a certain outcome
Modeling
Imitating a behavior seen in others
Vicarious learning
Learning to engage in a behavior or not, after seeing others being rewarded or punished for performing that action

Habituation

- A **decrease** in an innate response to a frequently repeated stimulus
  - Allows a better ability to focus on what is new by ignoring that which has already become familiar

*__dishabituation__ an increase in a response because of a *momentary* change in something familiar

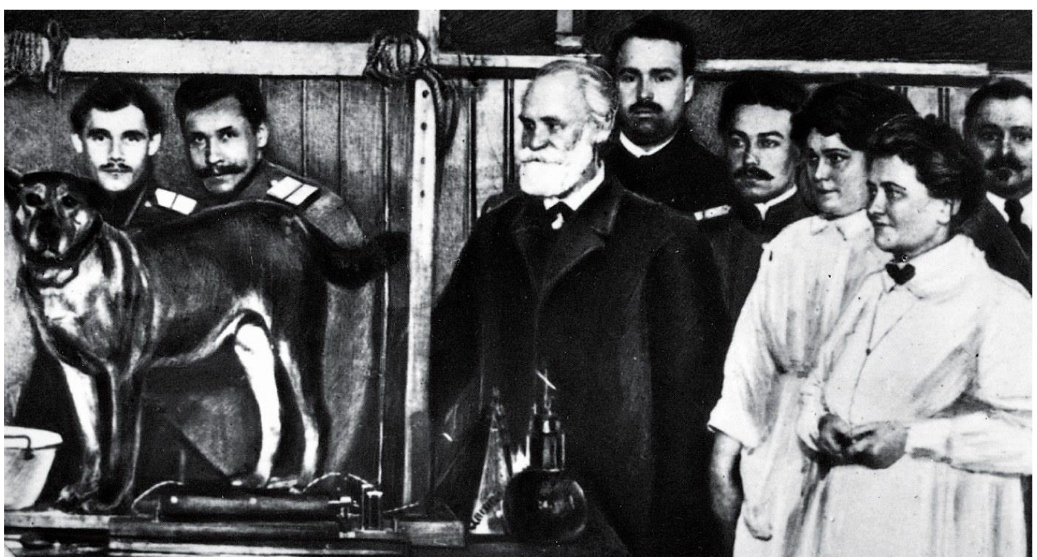
Sensitization

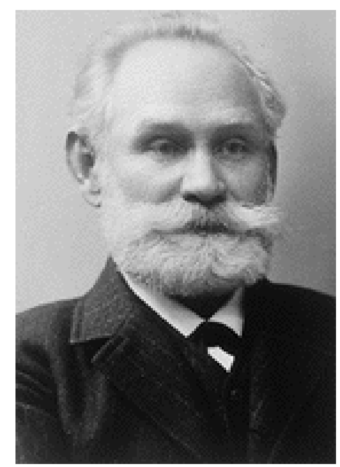
- An **increased** response to a stimulus after repeated exposure
  - It's a more generalized reactivity

*__desensitization__ is the extinction of sensitization

Associative learning: Classical conditioning

- A form of learning in which a neutral stimulus comes to elicit a response when it is associated with a stimulus that already produces that response





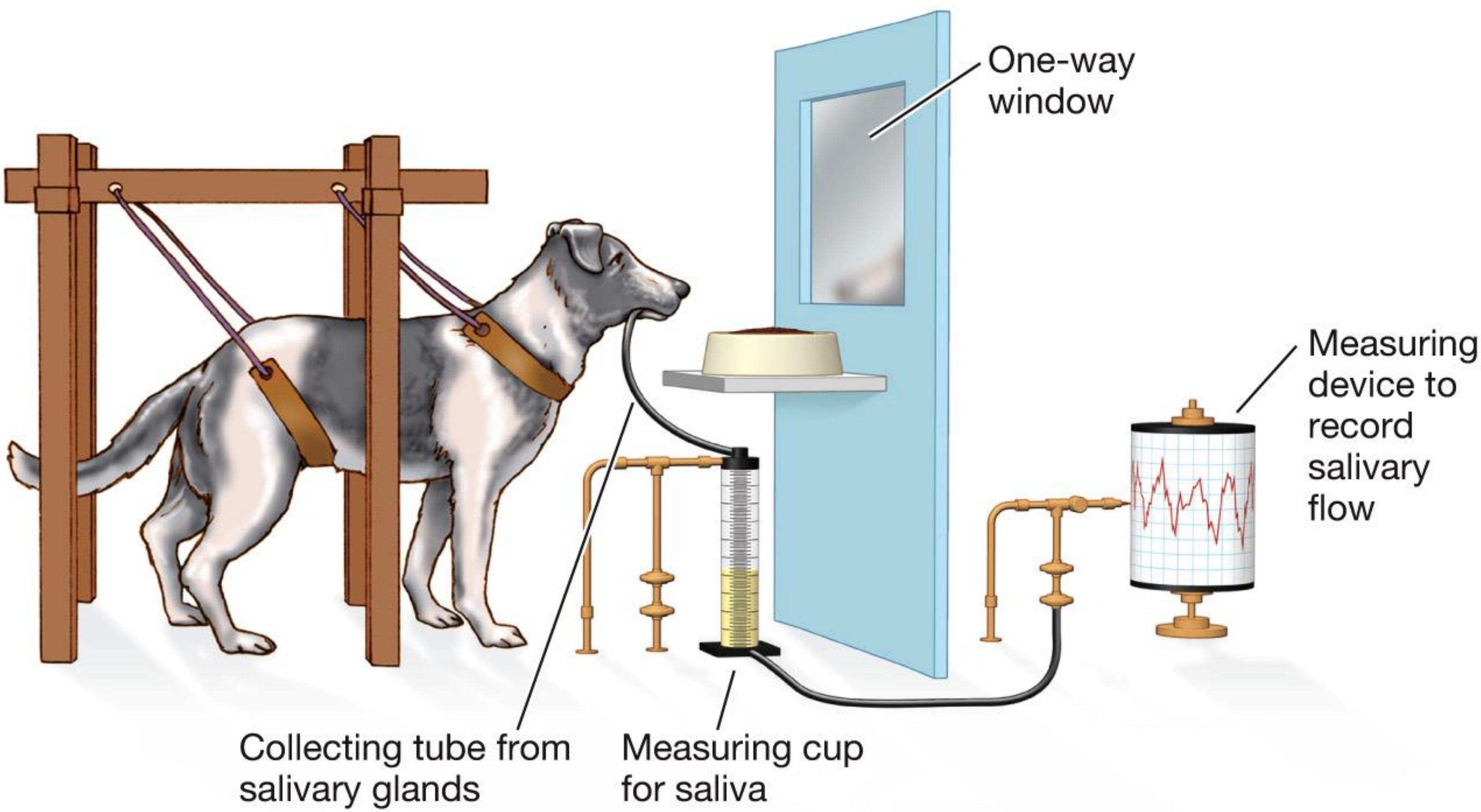
Pavlov, 1849-1936

Associative learning: Classical conditioning

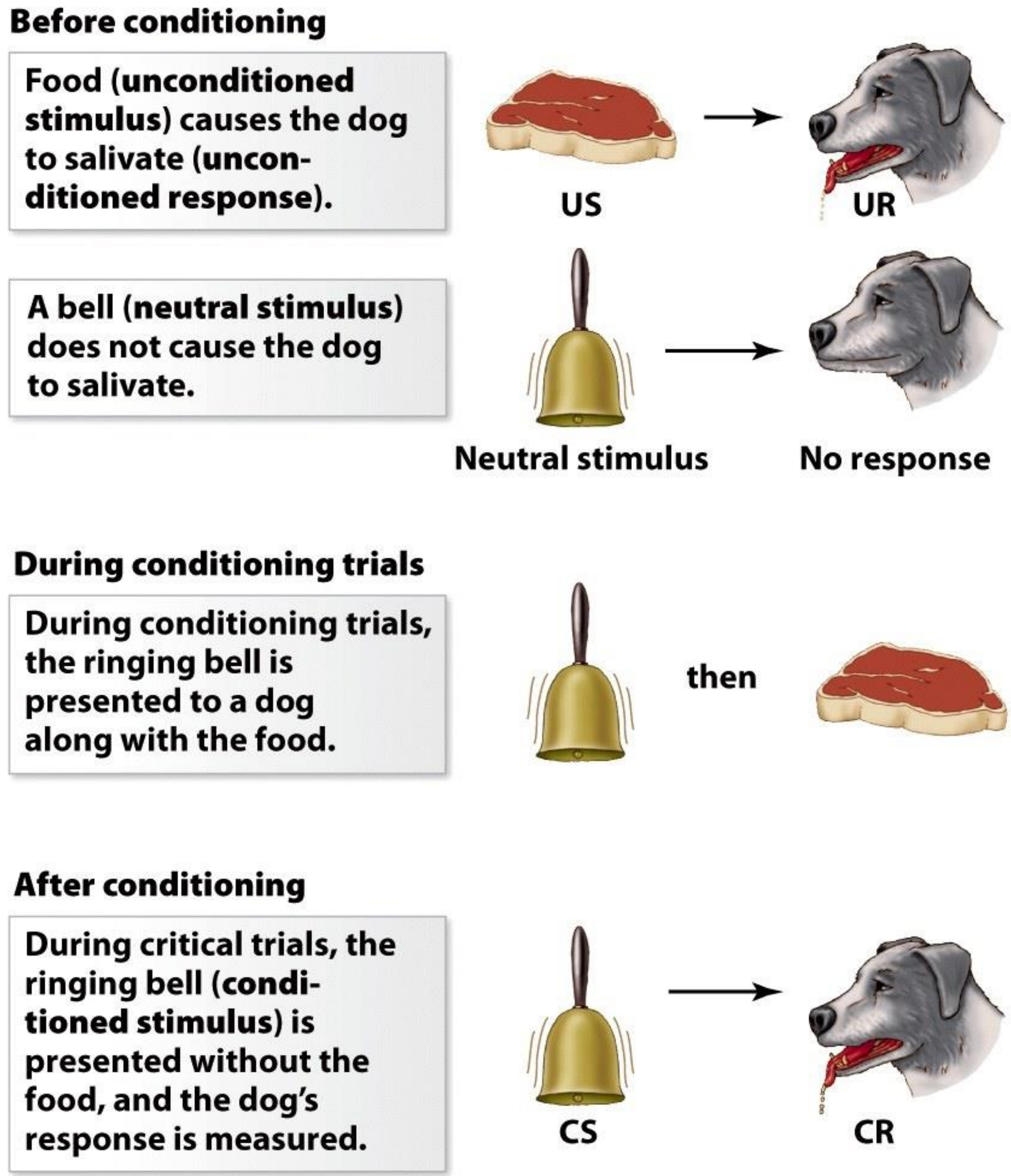
1 The dog is presented with a bowl that contains meat.

2 A tube carries the dog's saliva from the salivary glands to a container.

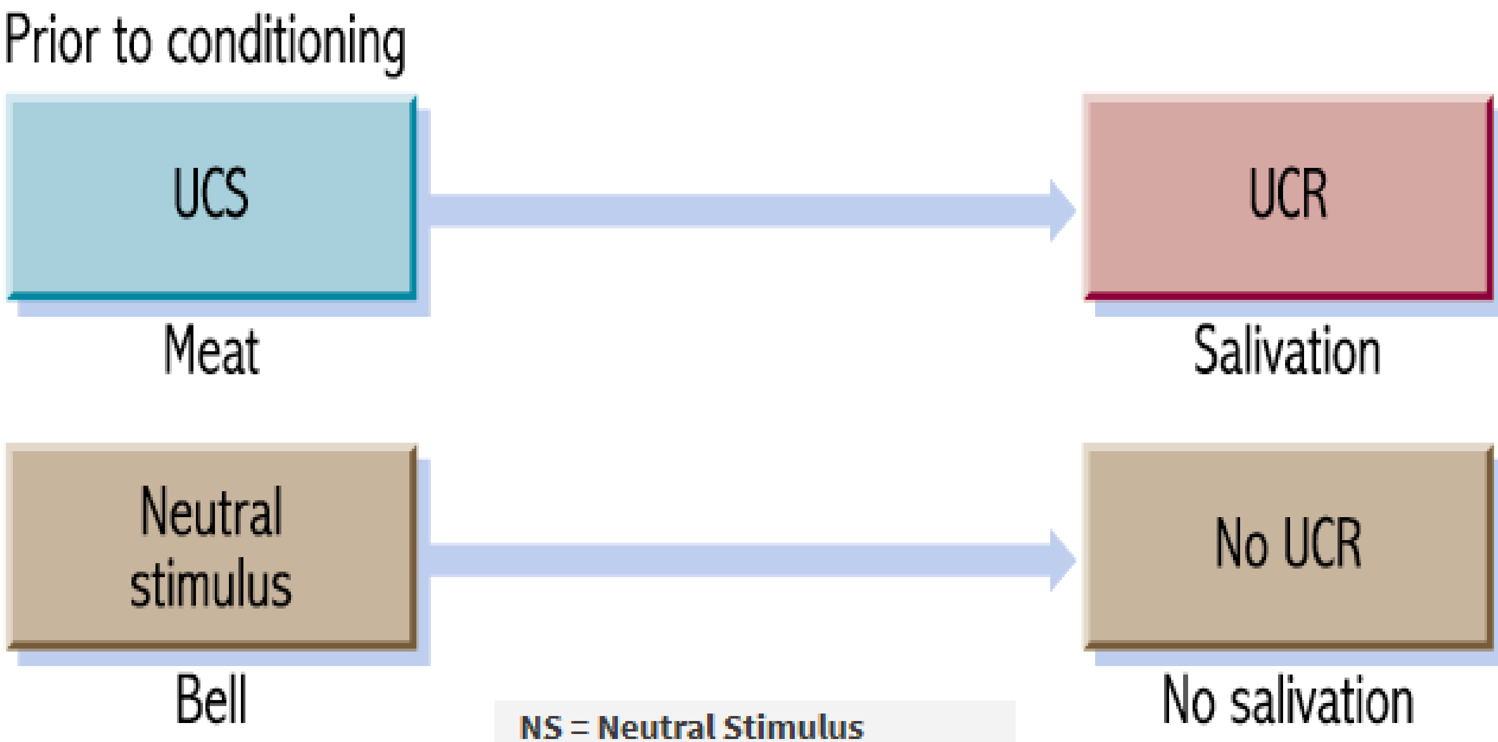
3 The container is connected to a device that measures the amount of saliva.



Associative learning: Classical conditioning

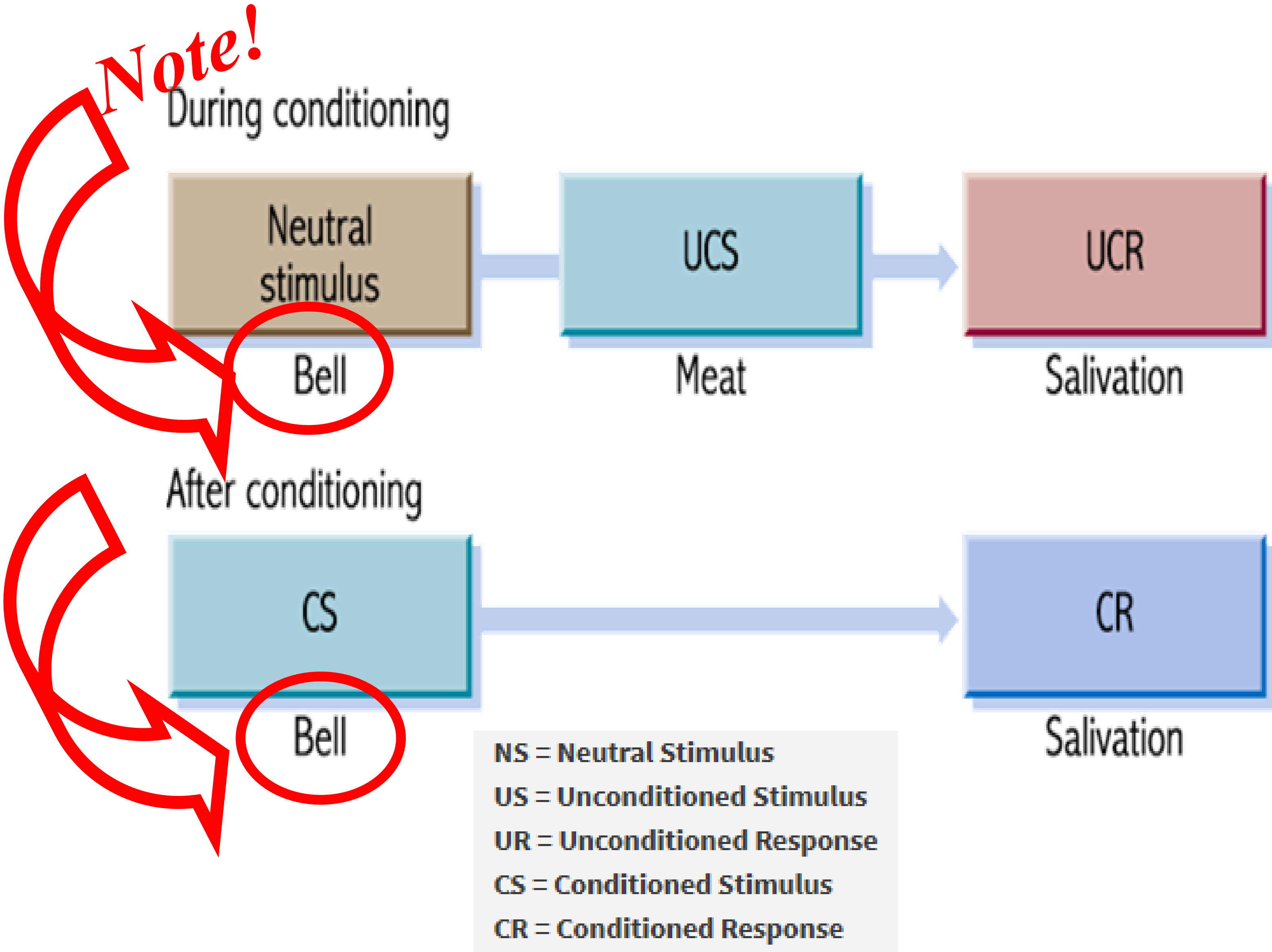


Conditioning Process

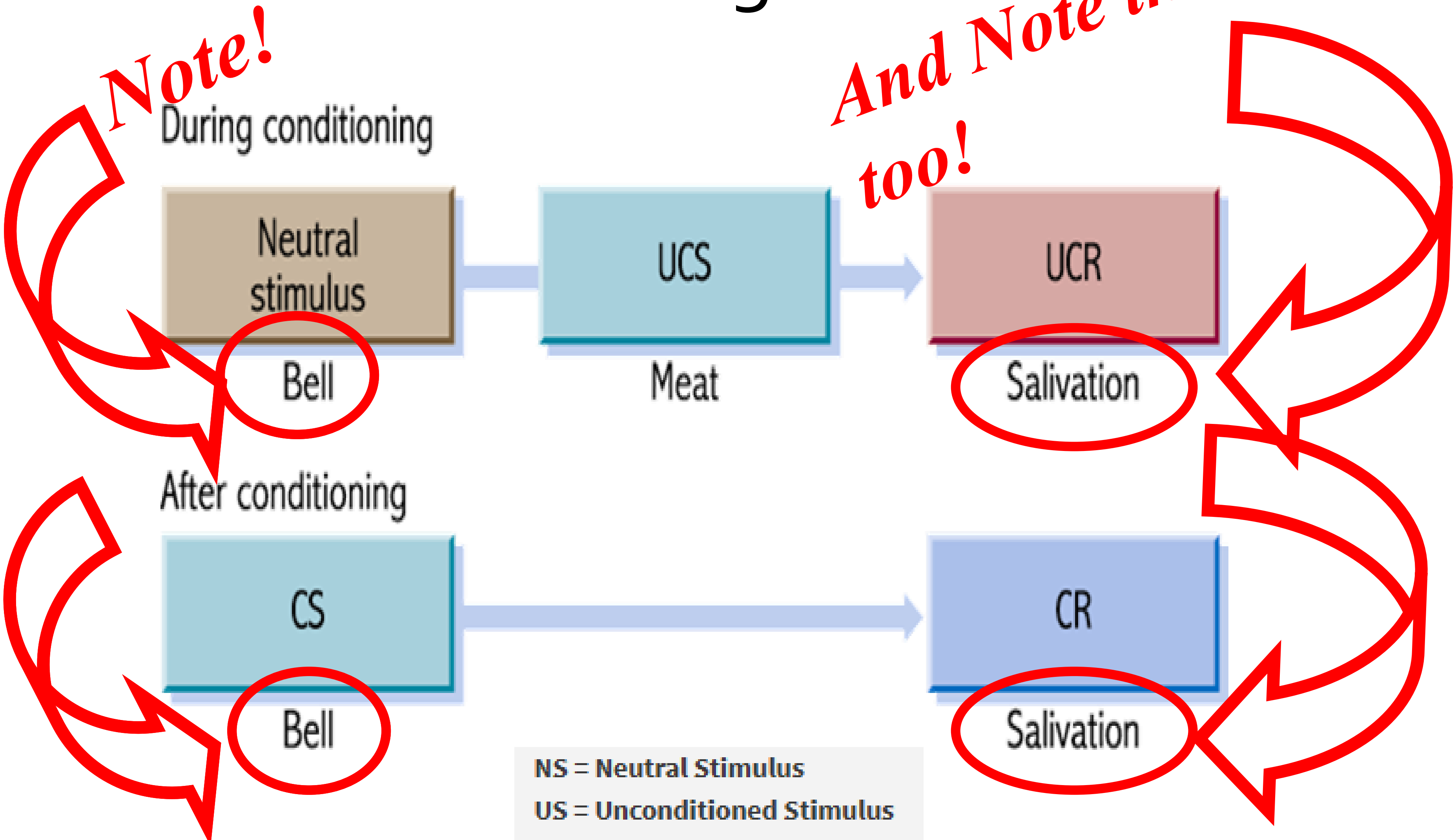


NS = Neutral Stimulus
US = Unconditioned Stimulus
UR = Unconditioned Response
CS = Conditioned Stimulus
CR = Conditioned Response

Conditioning Process



Conditioning Process



NS = Neutral Stimulus
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Associative learning: Classical conditioning

BEFORE TRAINING

US (food in mouth) → UR (salivation)

CS (e.g., tone) → No relevant response

Conditioning trials or reinforced trials

TRAINING

CS (tone) + US (food in mouth)

Critical trials or unreinforced trials

AFTER TRAINING (that is, conditioning)

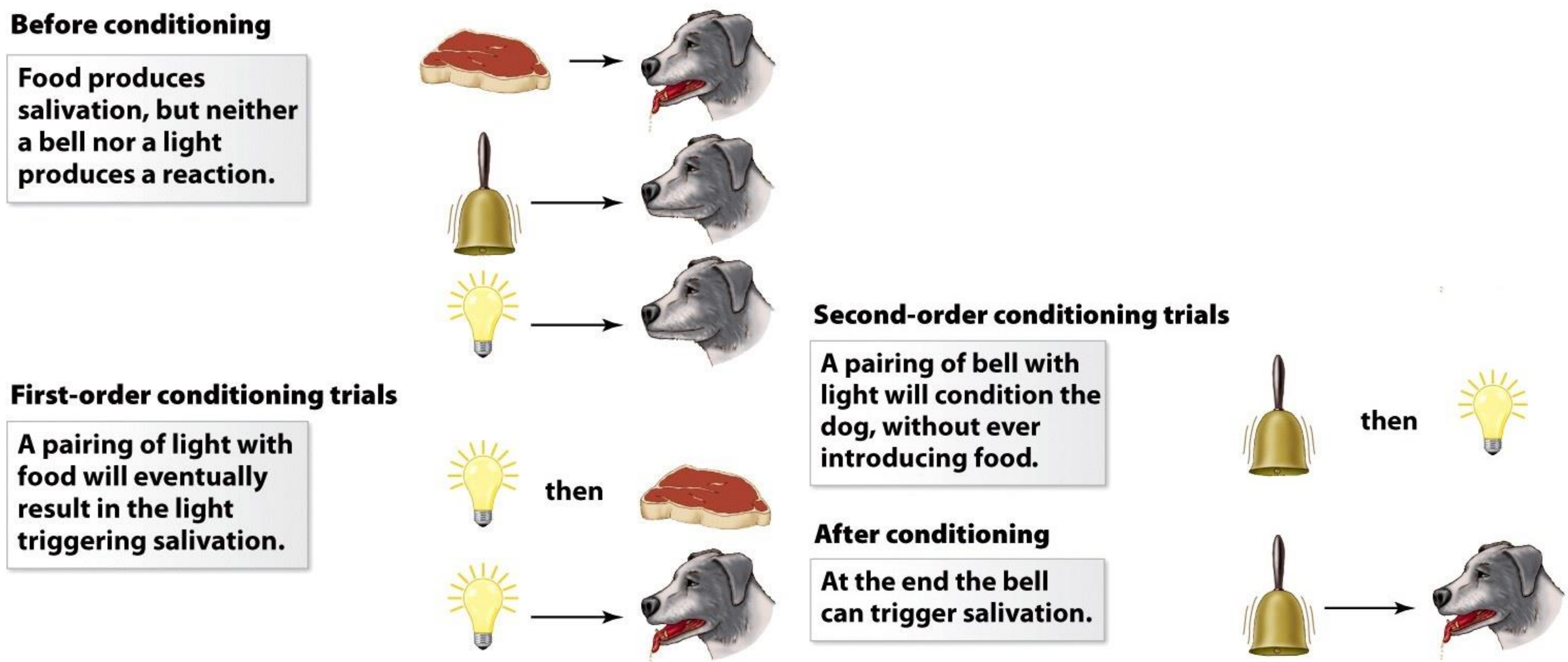
US (food in mouth) → UR (salivation)

CS (tone) → CR (salivation)

1 week of petting (CS)
while treats (US)

Associative learning: Classical conditioning

- Second-order conditioning
  - when a CS is paired with a new S, the new S produces CR



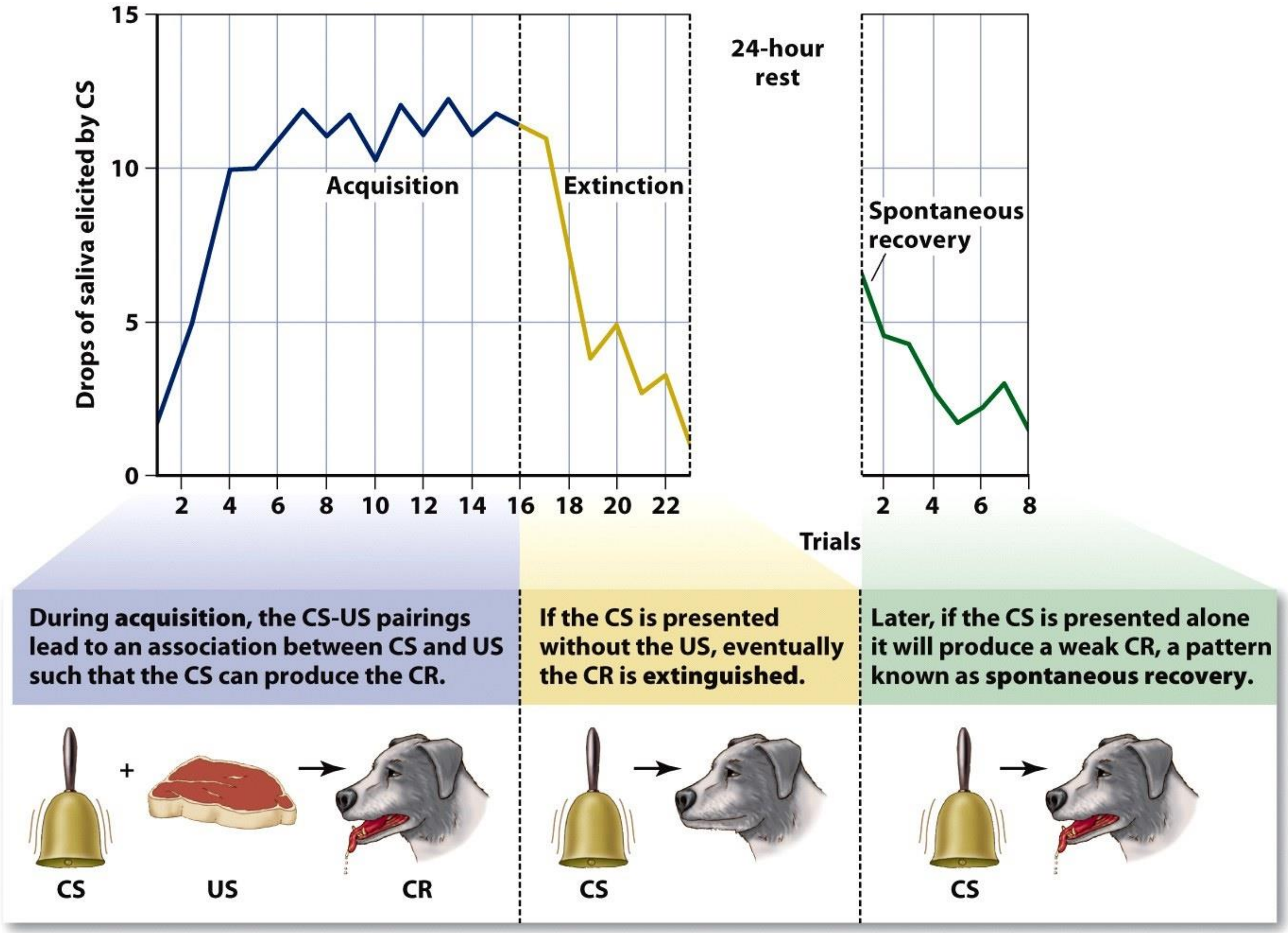
Associative learning: Classical conditioning

- Second-order conditioning
  - when a CS is paired with a new S, the new S produces CR

- Extinction
  - the association between the CS and CR can be eliminated by repeatedly presenting the CS alone.

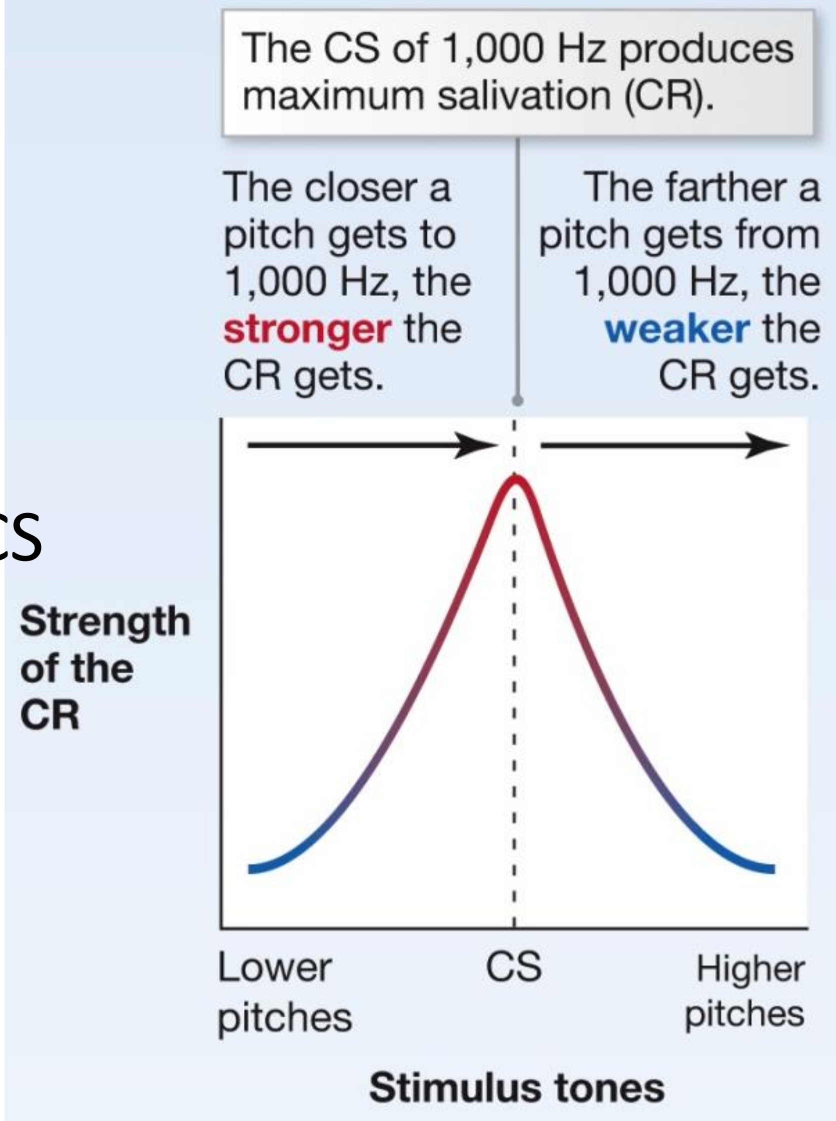
- Spontaneous recovery
  - after extinction and a resting interval

Associative learning: Classical conditioning



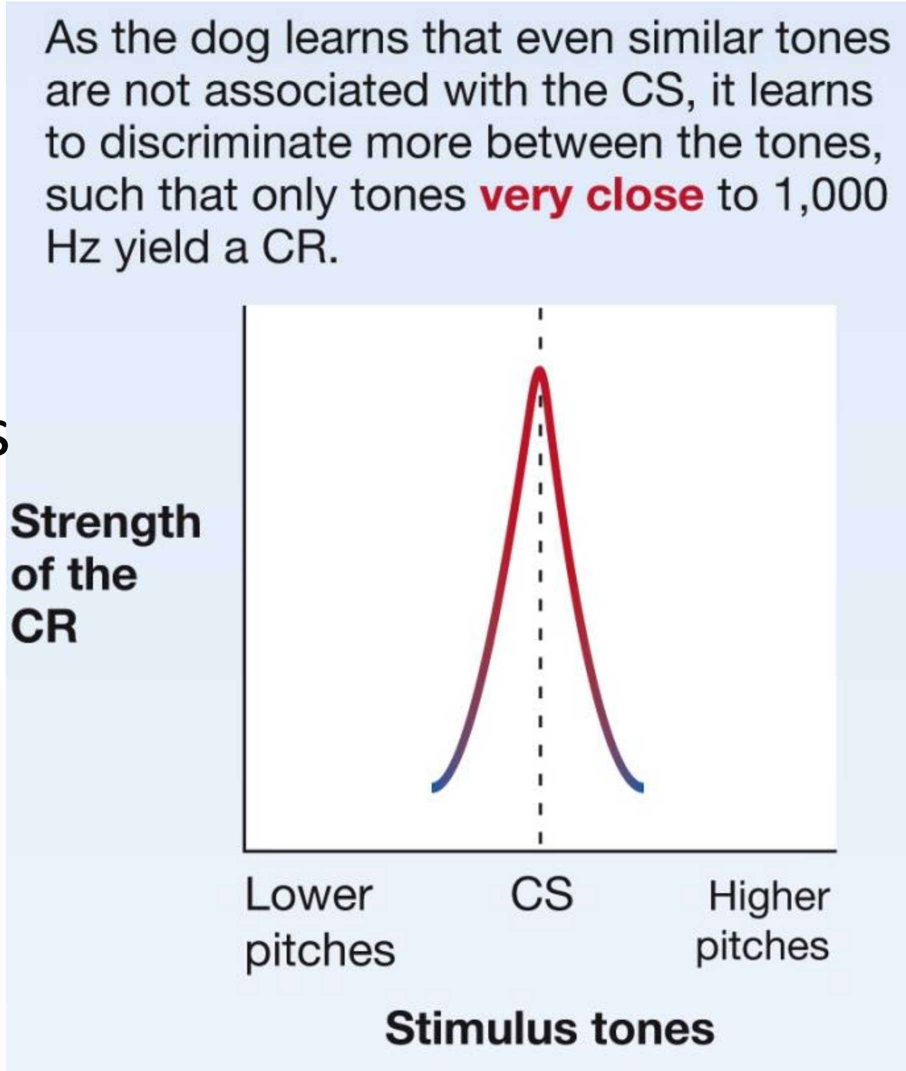
Associative learning: Classical conditioning

- Stimulus generalization
  - the CR can also be elicited by stimuli that are similar to the CS

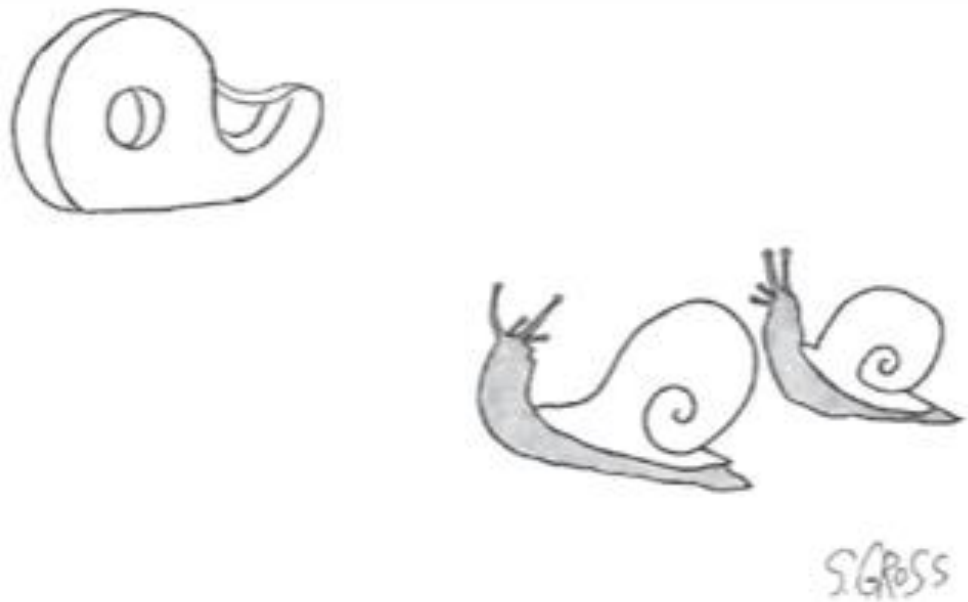


Associative learning: Classical conditioning

- Discrimination
  - CS+ -> reinforced stimulus
  - CS- -> unreinforced stimulus



What conditioning principle is influencing the snail's affections?



"I don't care if she's a tape dispenser.
I love her."

Myers/DeWall, *Psychology*, 13e, ©
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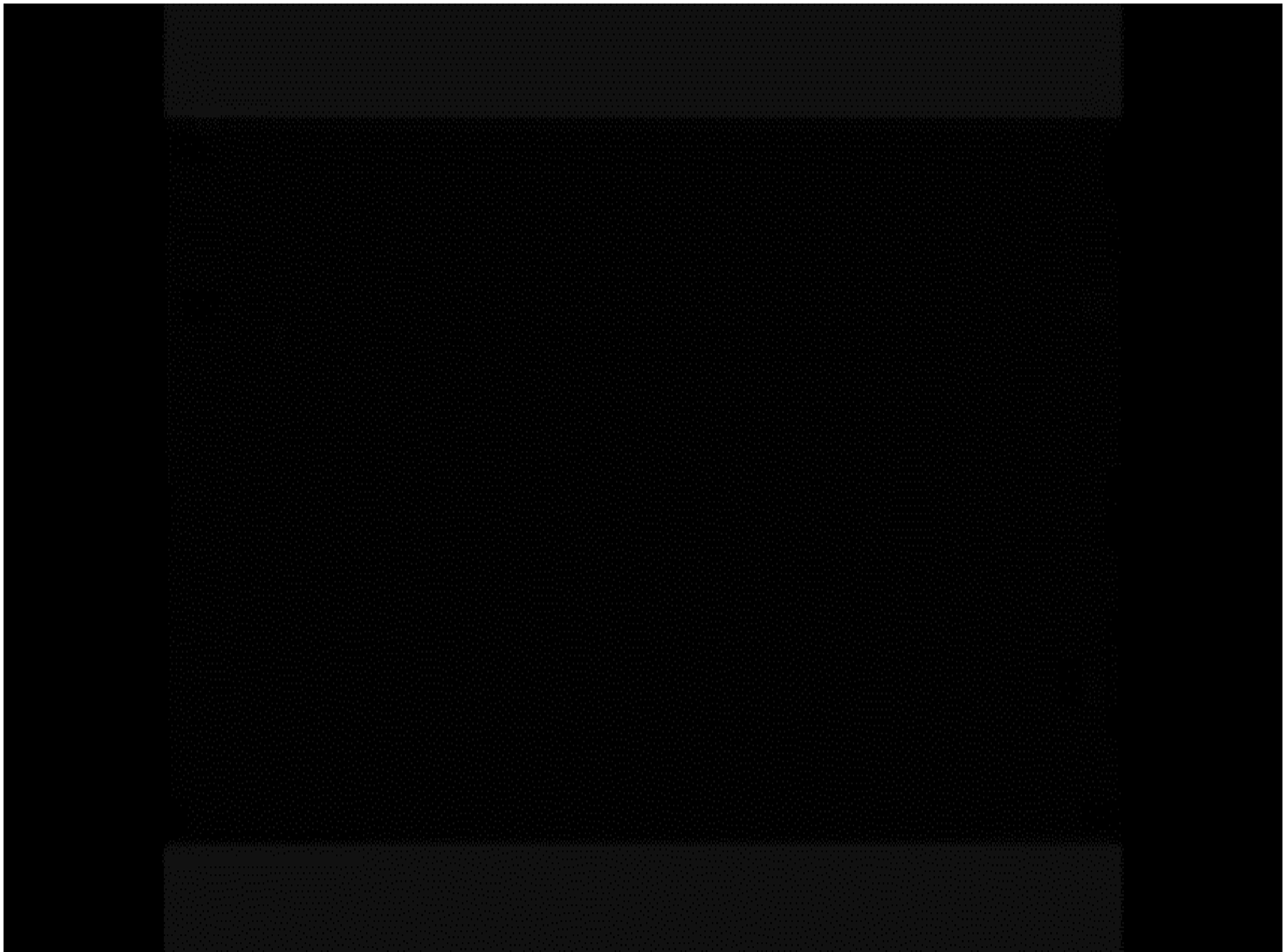
Associative learning: Classical conditioning



Applications of Classical Conditioning

NS = Neutral Stimulus
US = Unconditioned Stimulus
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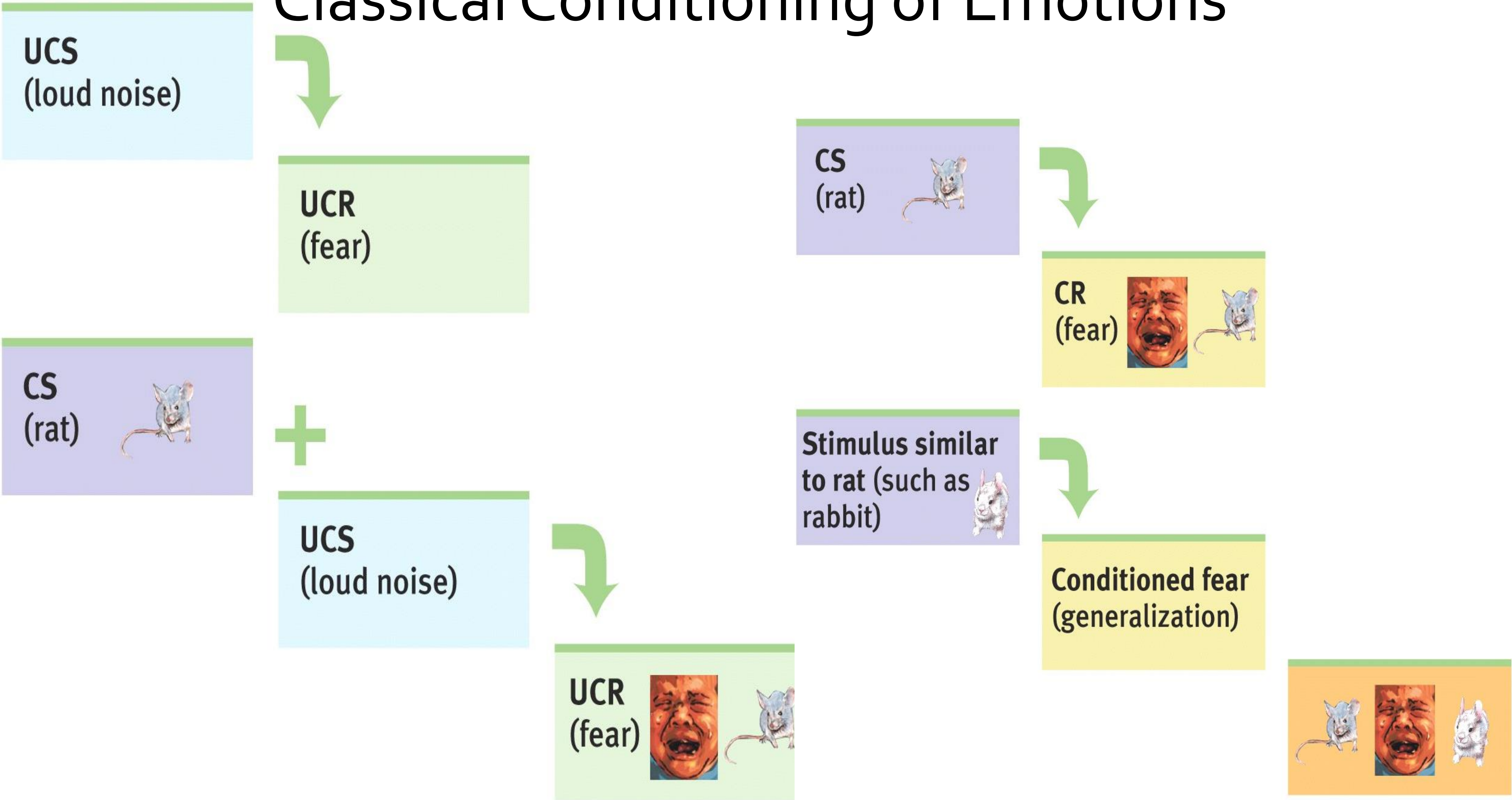
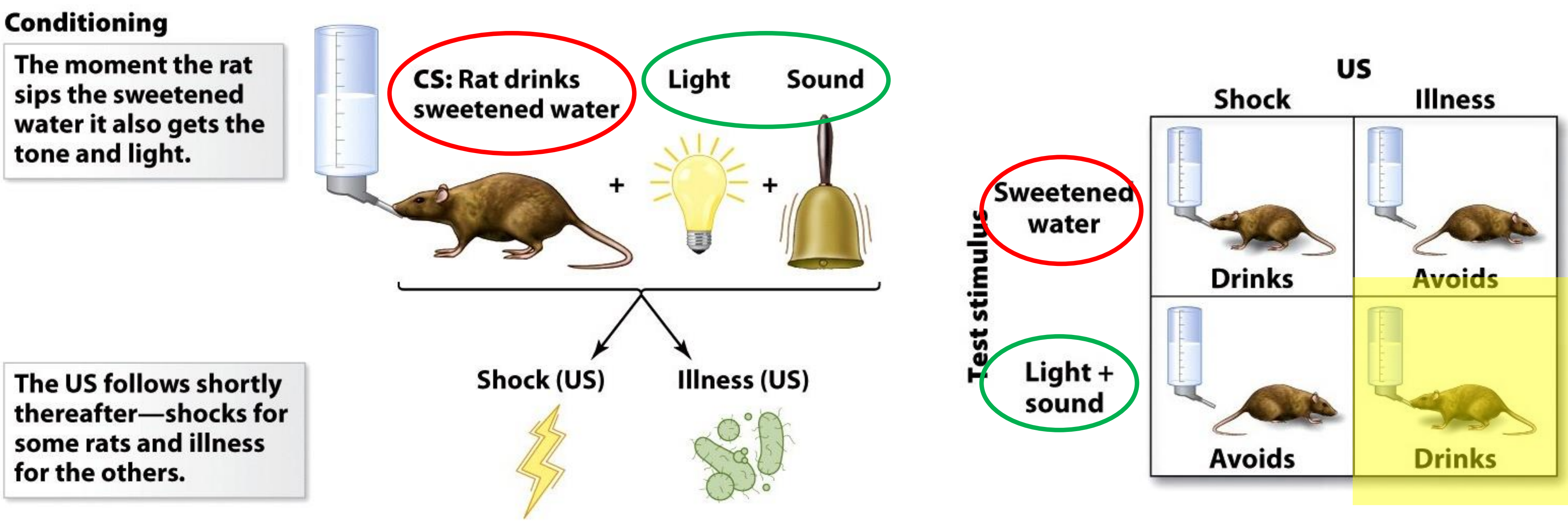
Classical Conditioning of Emotions



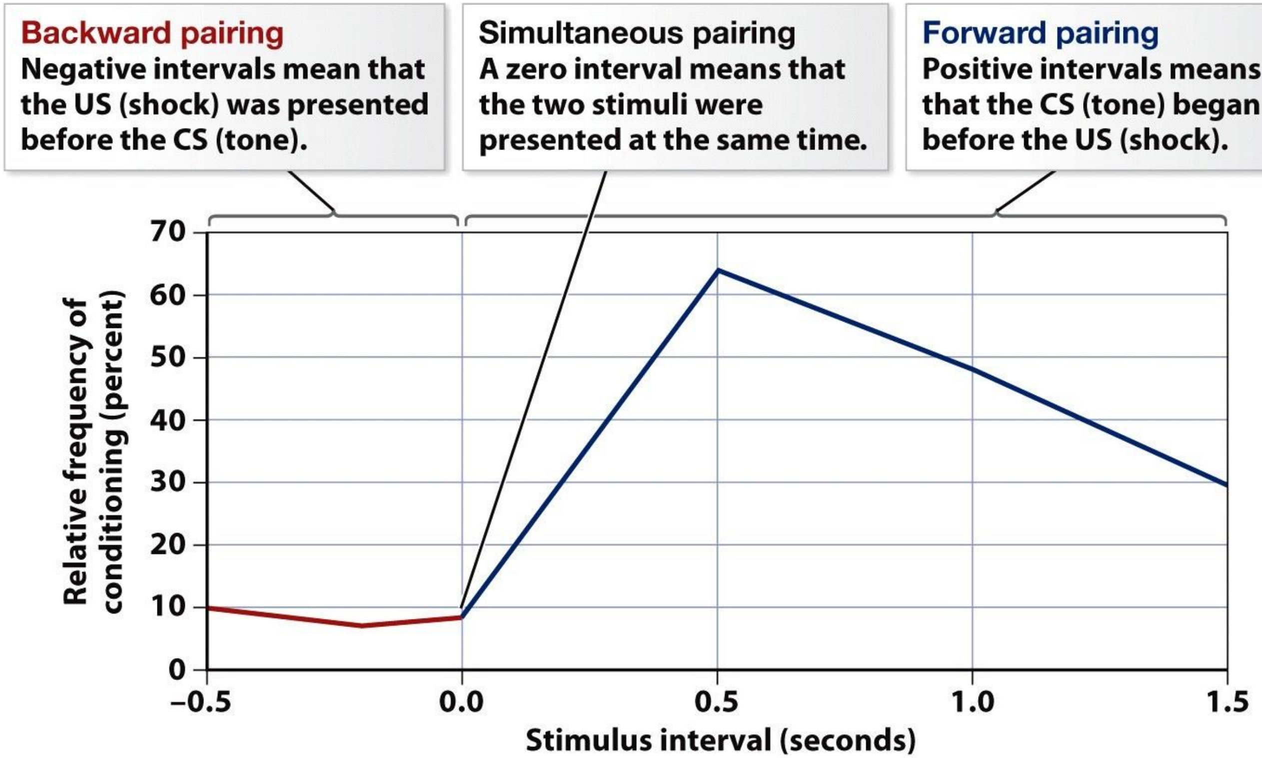
Associative learning: Classical conditioning

- BUT an organism cannot learn **every** association
  - Biological constraints → biological preparedness



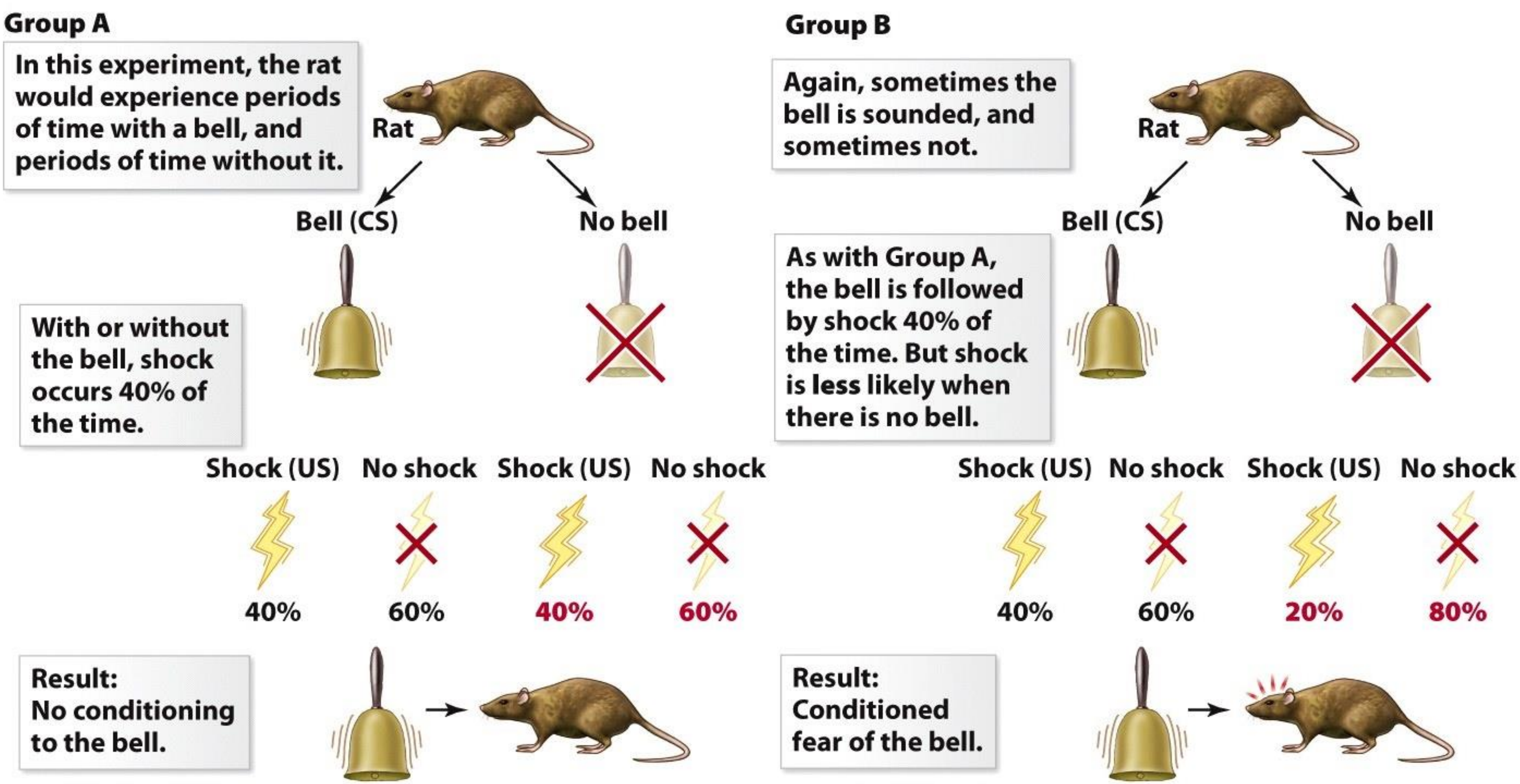
Associative learning: Classical conditioning

- BUT → conditioning requires
  - **Contiguity** (together in time) – such that learning only occurs when the CS is presented **before** the US



Associative learning: Classical conditioning

- BUT → conditioning requires
  - **Contingency** – the CS should be informative with respect to the chance that the US is offered



Associative learning: Classical conditioning

- Cognitivist view:
  - The CS functions as a sort of *predictor* of the US!
  - The CR is a reaction in preparation of the US.

Associative learning: Classical conditioning

- This can also lead to compensatory reactions
  - Drugs

INITIAL STATE

Morphine or heroine → Euphoria
Decreased pain sensitivity

AFTER LEARNING: DRUG TOLERANCE

Morphine or heroine → { Euphoria
Decreased pain sensitivity
etc. }

\+

Conditioned stimuli → { Dysphoria
Increased pain sensitivity
etc. }

→ No change

NO CRAVING

Conditioned stimuli
(with no drug) → { Dysphoria
Increased pain sensitivity
etc. }

Applications of Classical Conditioning

- This can also lead to compensatory reactions
  - Drugs
    - Former crack cocaine users avoid cues (people, places) associated with previous drug use.
  - Immune responses
  - Food cravings

Applications of Classical Conditioning

NS = Neutral Stimulus
US = Unconditioned Stimulus
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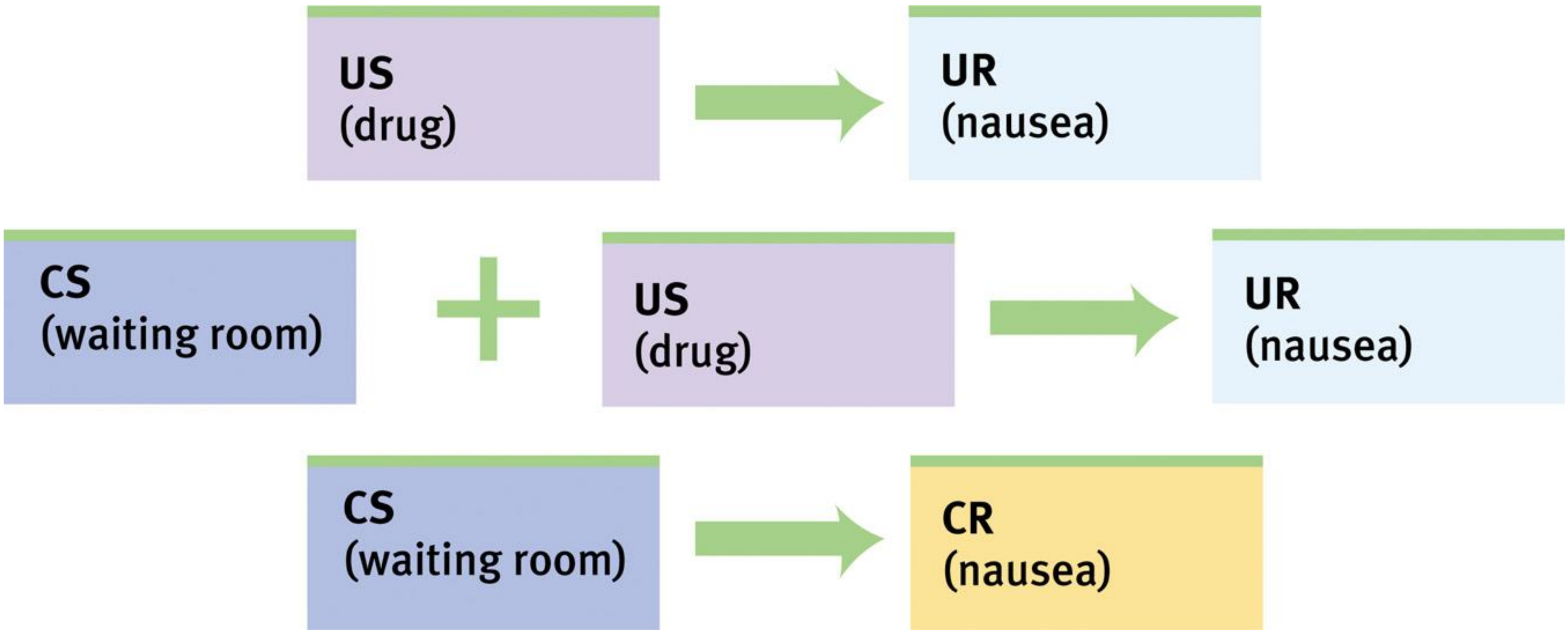
CS = Conditioned Stimulus
CR = Conditioned Response

Nausea Conditioning in Cancer Patients



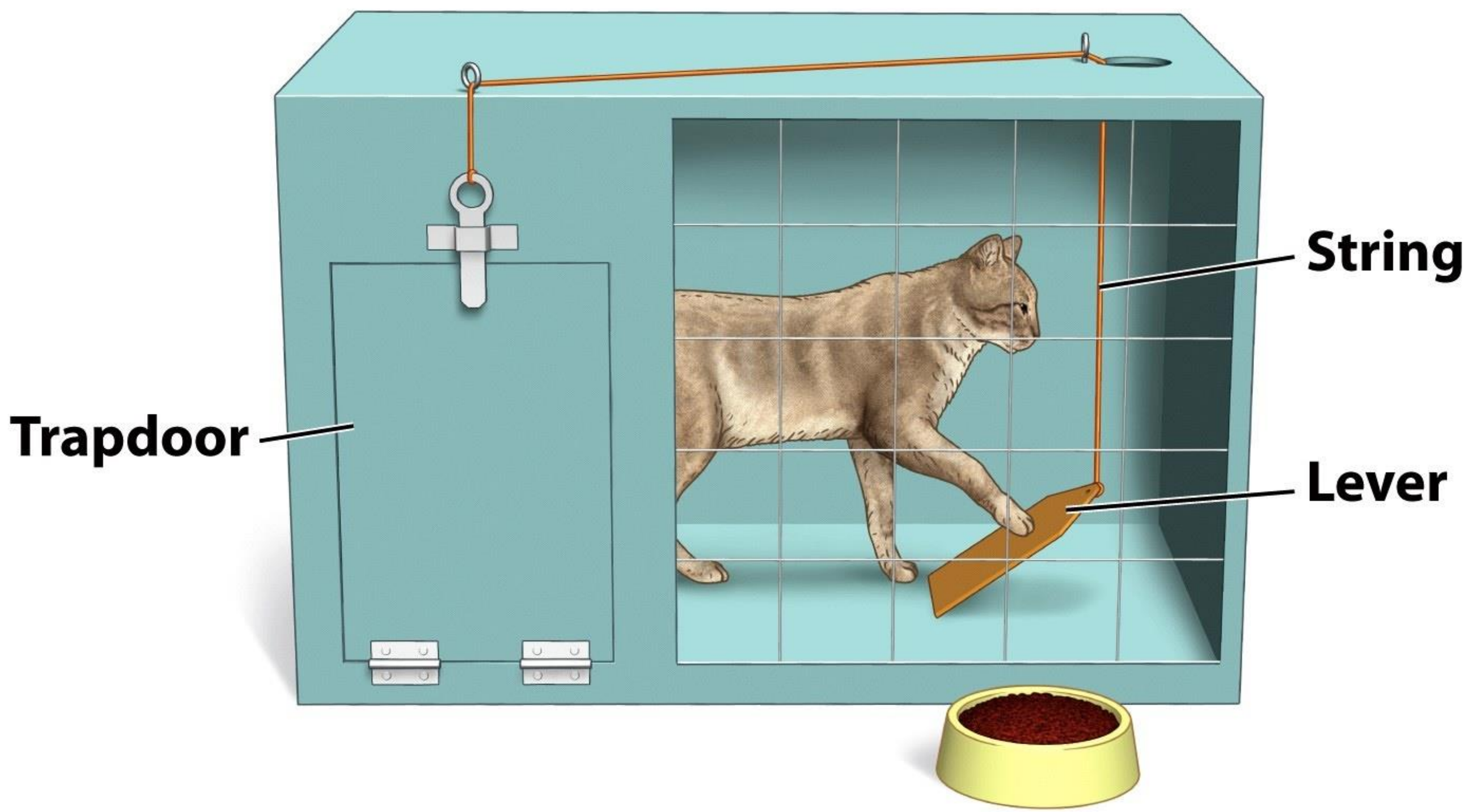
Associative learning: Operant conditioning

- *a.k.a.* instrumental conditioning
- The *consequence* of an action/response determines the likelihood that it will be performed again.
- Learned is the relationship between response and the consequence (reinforcer/reward or punishment).

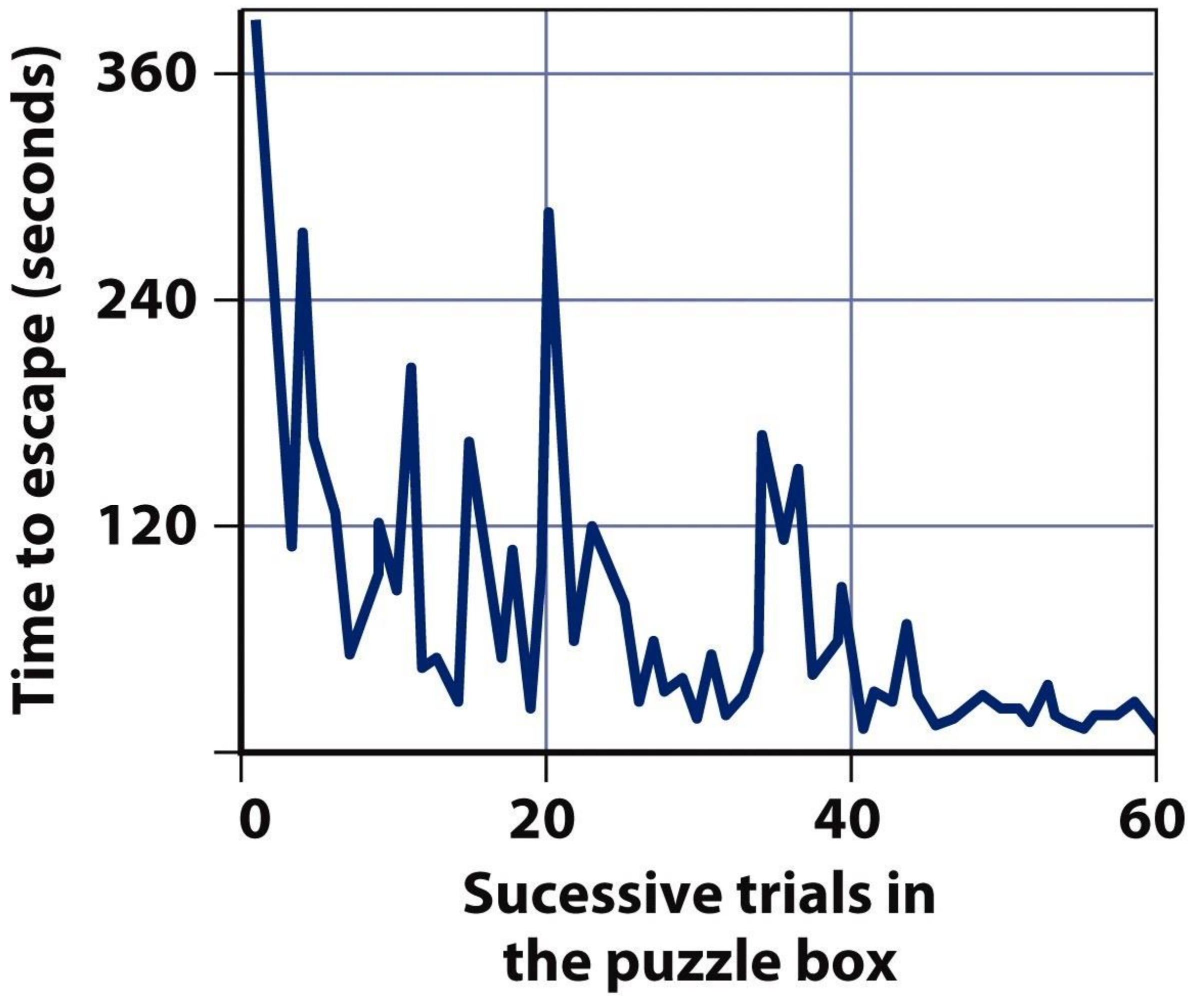
Associative learning: Operant conditioning

- *Law of Effect* (Thorndike, 1898)
  - Behavior is governed by its consequences.
  - Performance is strengthened if it’s followed by a reward and weakened if it’s not.
    - i.e., the tendency to give a response

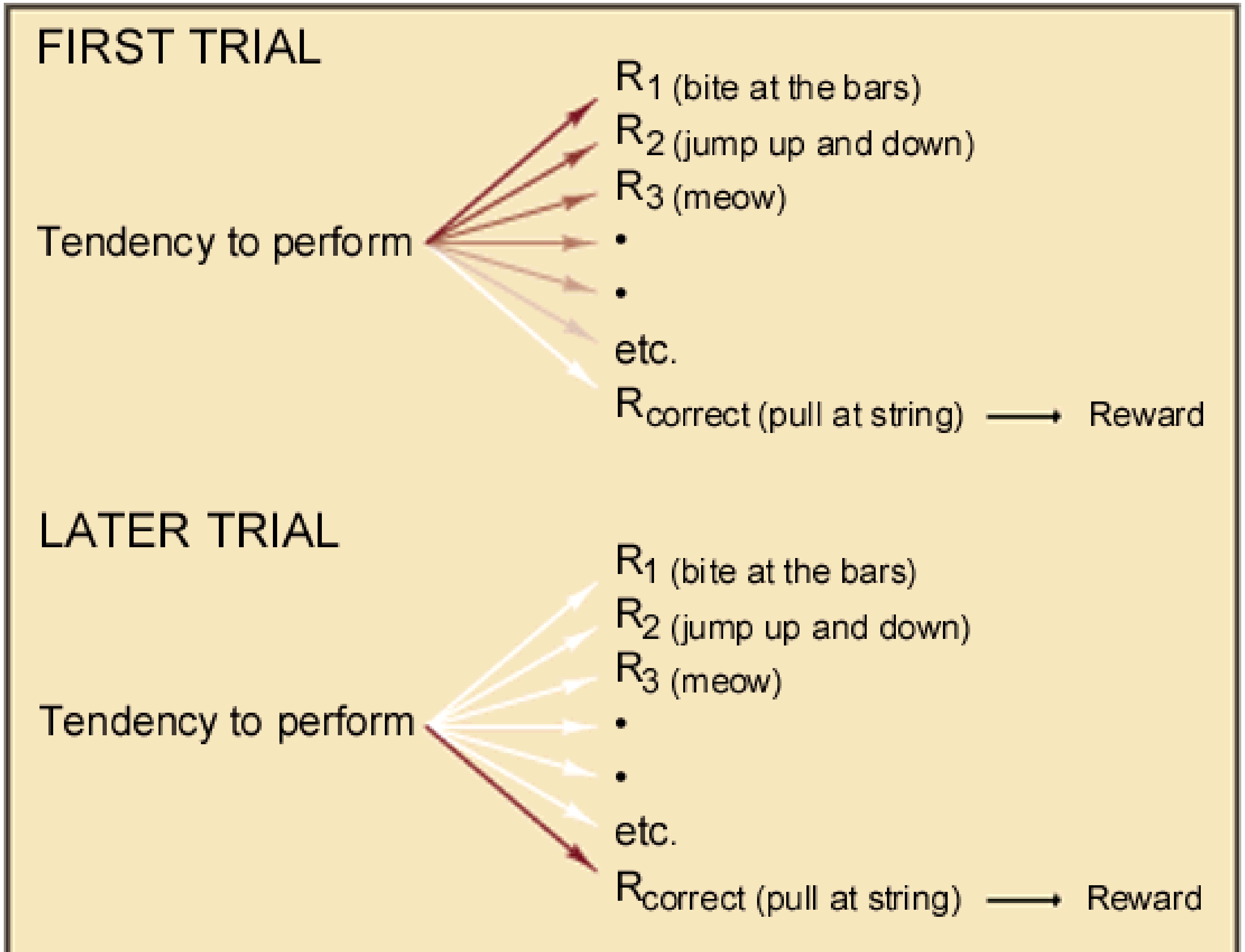
Associative learning: Operant conditioning



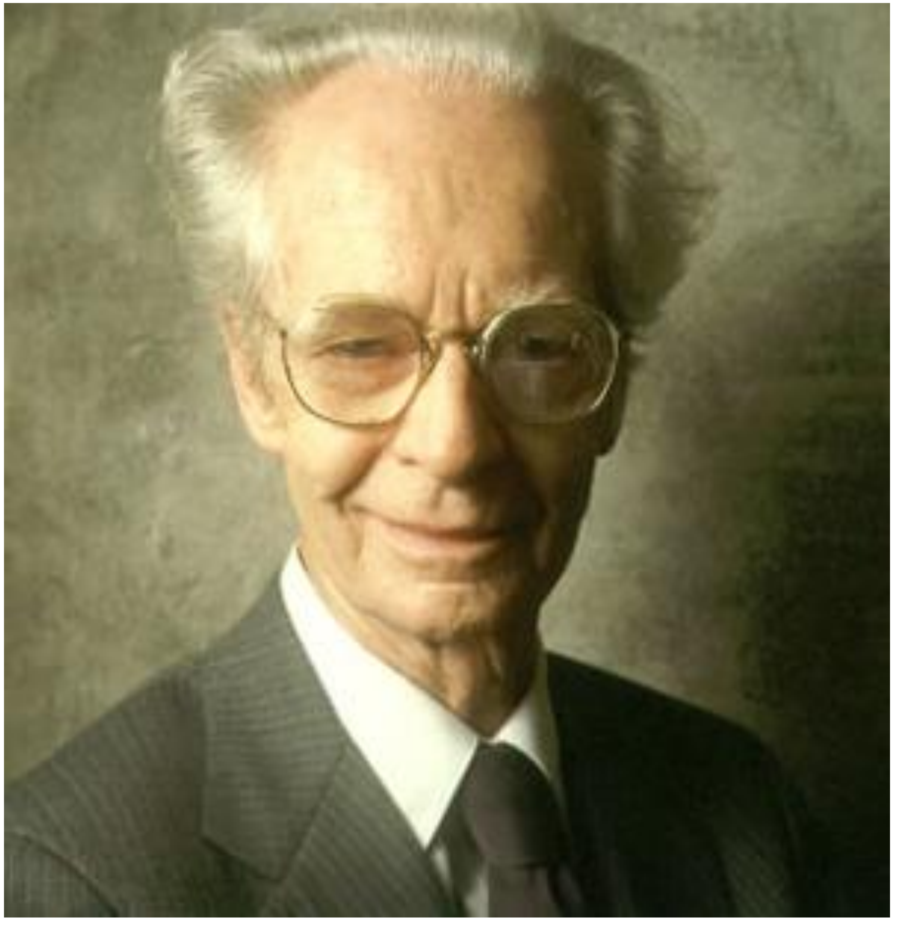
Associative learning: Operant conditioning



Associative learning: Operant conditioning

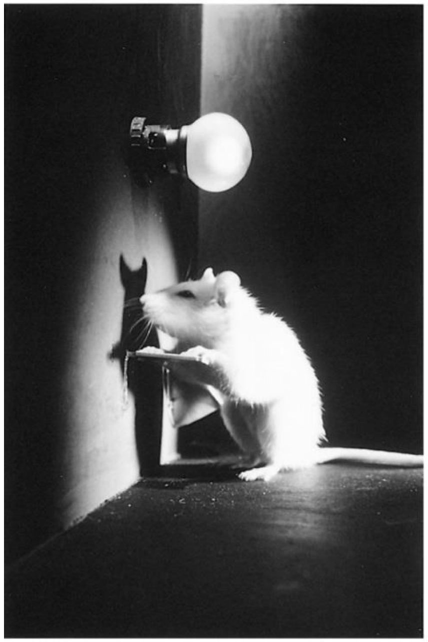


Associative learning: Operant conditioning



B.F. Skinner (1904-1990)

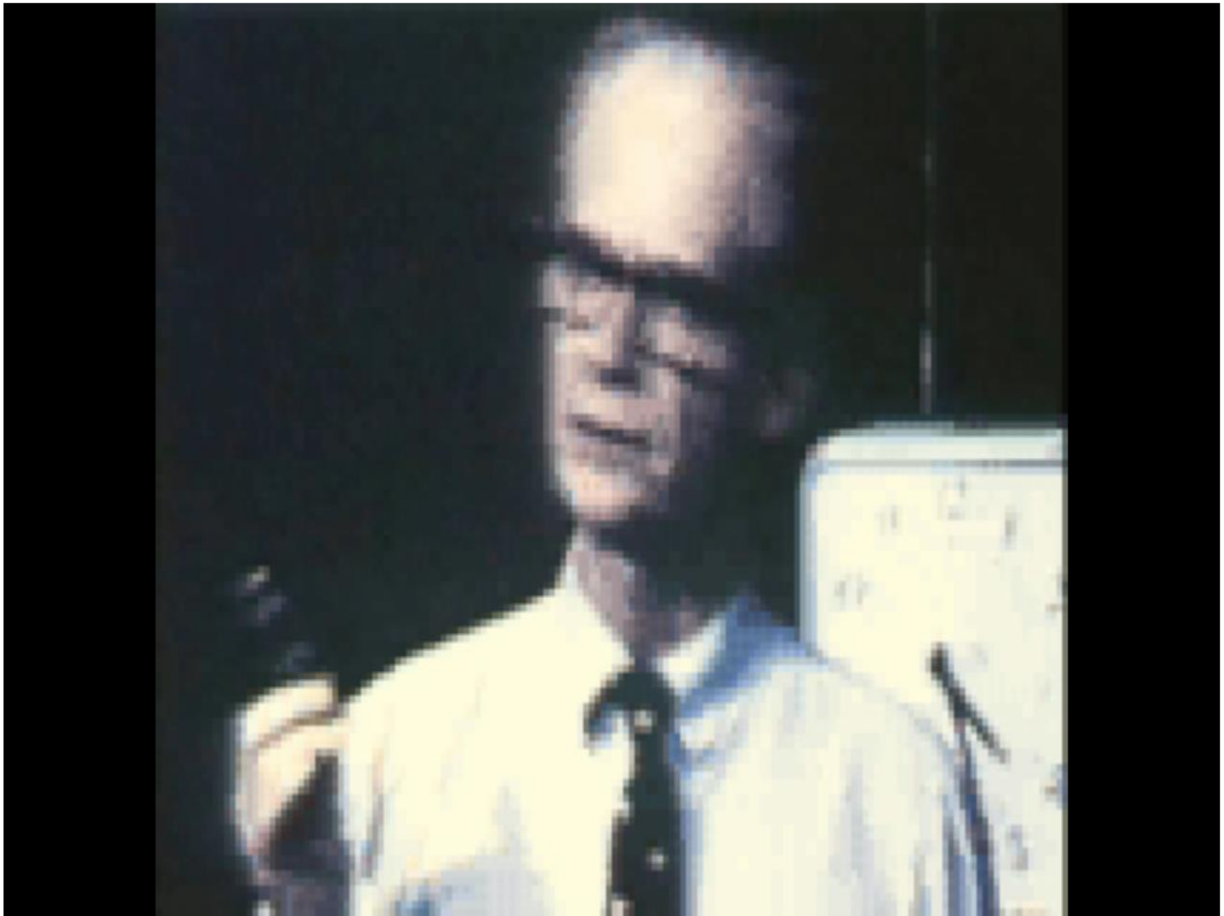
Associative learning: Operant conditioning



Operant chambers or Skinner boxes

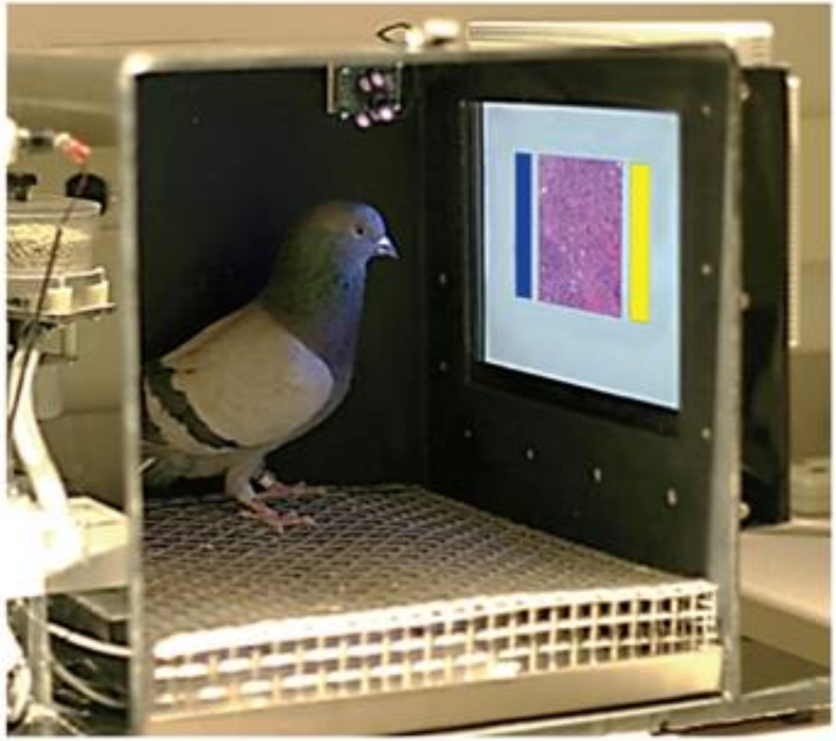
Associative learning: Operant conditioning

- Shaping
  - successive approximations



From 6th minute: pigeon superstition

https://www.youtube.com/watch?v=6XWBXQAa-IU&ab_channel=TheFaculties



Levenson RM, Krupinski EA, Navarro VM, Wasserman EA (2015) Pigeons (*Columba livia*) as Trainable Observers of Pathology and Radiology Breast Cancer Images. PLoS ONE 10(11): e0141357.

Bird brains spot tumors After being rewarded with food when correctly spotting breast tumors, pigeons became as skilled as humans at discriminating cancerous from healthy tissue (Levenson et al., 2015). Other animals have been shaped to sniff out land mines or locate people amid rubble (La Londe et al., 2015).

How is operant conditioning at work in this cartoon?

Associative learning: Operant conditioning

- Shaping
  - successive approximations
- Reinforcers
  - Primary and secondary reinforces
  - Positive and negative reinforcers (removal of unpleasant)

Process	Behaviour	Consequence	Result
Reinforcement			
Positive reinforcement	Response occurs →	A stimulus is presented →	Response increases
	(Rat presses a lever)	(Food pellets appear)	(Lever pressing increases)
Negative reinforcement	Response occurs →	An aversive stimulus is removed →	Response increases
	(Person takes aspirin)	(Headache pain goes away)	(Increased tendency to take aspirin for headache relief)


cat tricks 101
@littlevoidluna

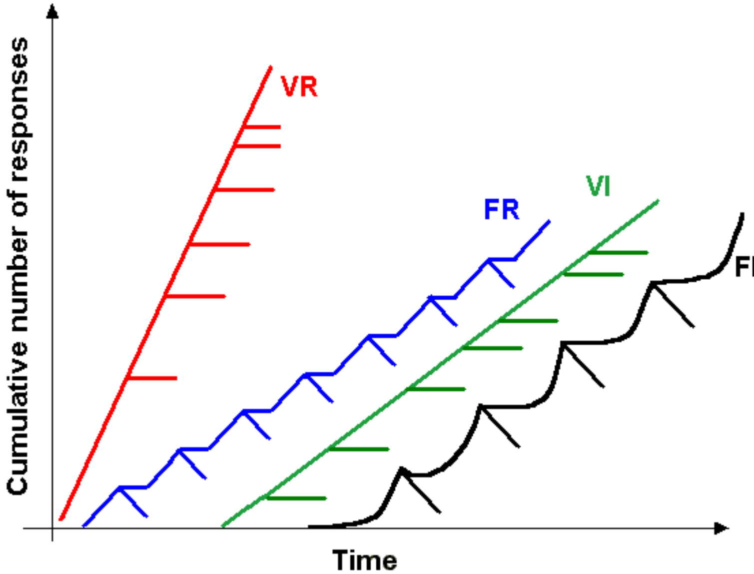
Associative learning: Operant conditioning

- Shaping
  - successive approximations
- Reinforcers
  - Primary and secondary reinforces
  - Positive and negative reinforcers (removal of unpleasant)
- Punishment
  - Positive and negative punishment (removal of pleasant)

Process	Behaviour	Consequence	Result
Punishment			
Aversive punishment (also called **positive punishment**)	Response occurs → (Two siblings fight over a toy)	An aversive stimulus is presented → (Parents scold or smack them)	Response decreases (Fighting decreases)
Response cost (also called **negative punishment**)	Response occurs → (Two siblings fight over a toy)	A stimulus is removed → (No TV for 1 week)	Response decreases (Fighting decreases)

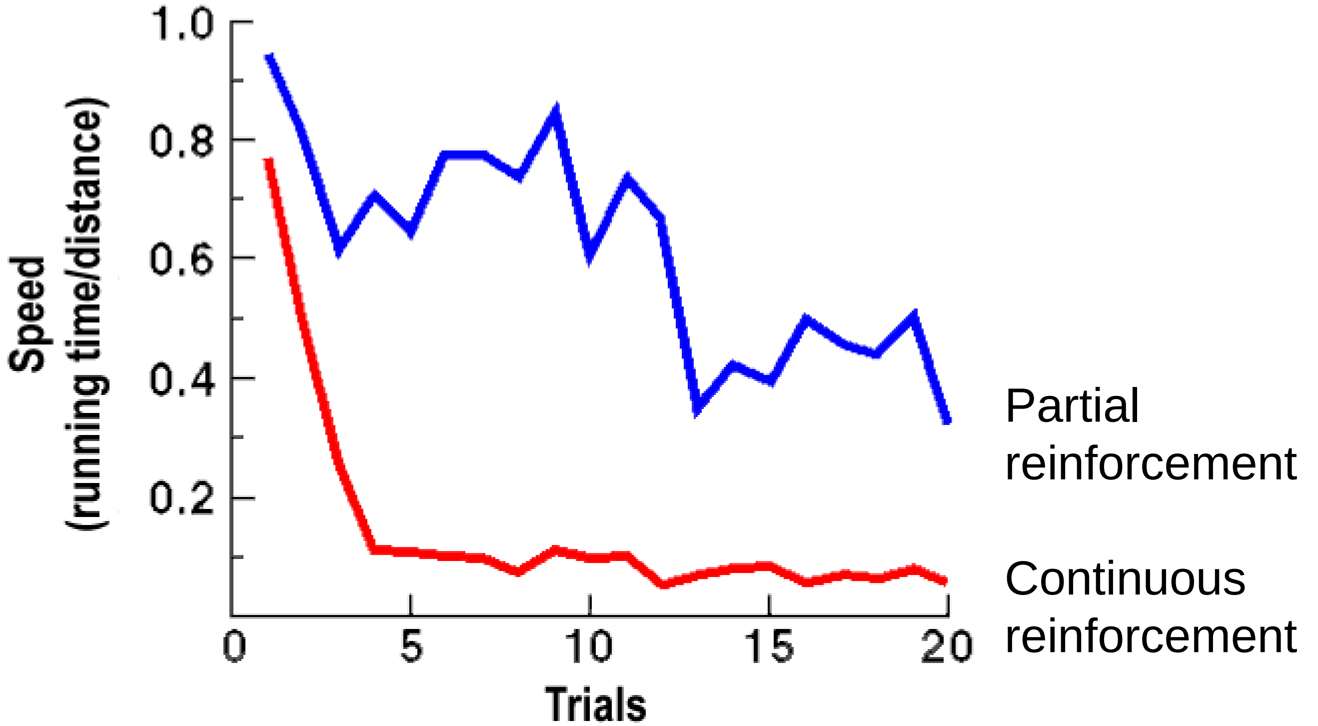
Associative learning: Operant conditioning

- Continuous reinforcement
- Partial reinforcement → Schedules of reinforcement
  - number
    - Fixed-Ratio (FR), e.g., food pellet after 4 responses
    - Variable-Ratio (VR), e.g., food pellet after on average 4 responses
  - time
    - Fixed-Interval (FI), e.g., food pellet after 1st response after 4 minutes
    - Variable-Interval (VI), e.g., food pellet after 1st response after on average 4 minutes



Associative learning: Operant conditioning

- Partial-reinforcement extinction effect
  - Partial reinforcement leads to less extinction than continuous reinforcement.



Associative learning: Operant conditioning

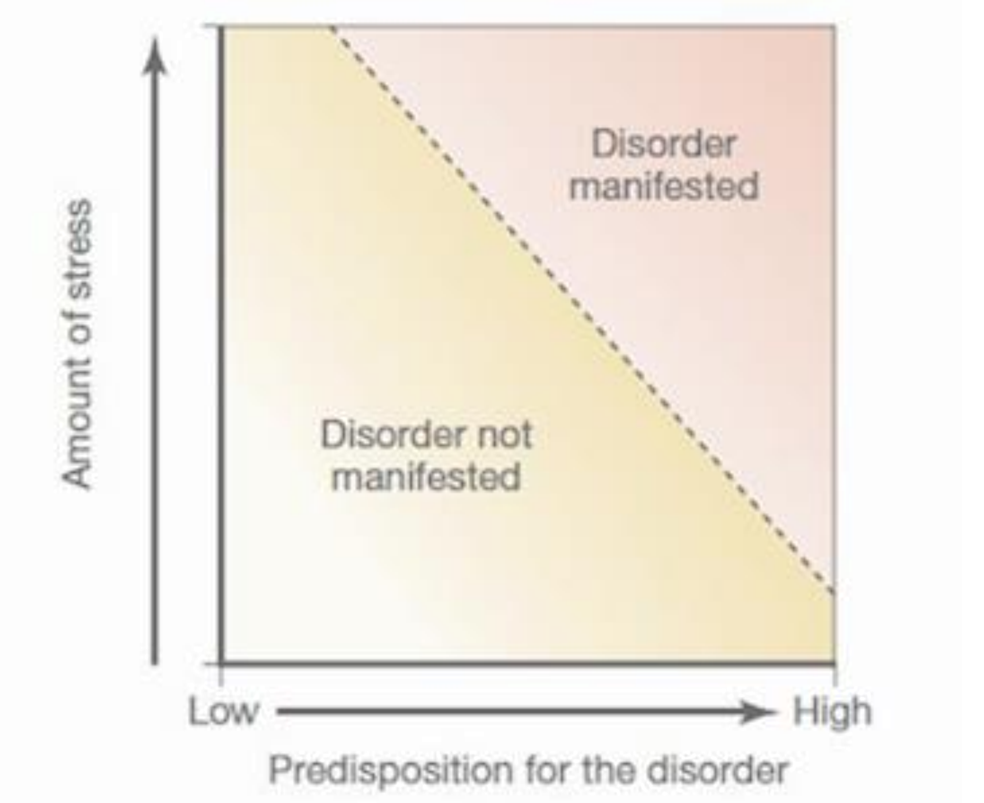
- BUT an organism cannot learn any association
  - Biological constraints ⟶ built-in predispositions

Heinlein (1907–1988) said it well:

“Never try to teach a pig to sing; it wastes your time and annoys the pig.”

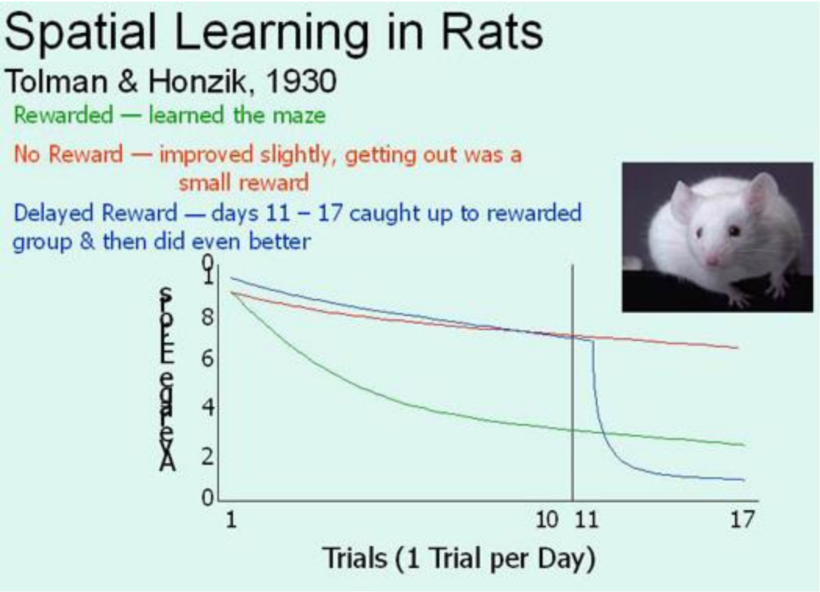
“Give me a dozen healthy infants, well formed, and my own specified world to bring them up in and I’ll guarantee to take any one at random and train him to become any type of specialist I might select—doctor, lawyer, artist, merchant-chief, and yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations and race of his ancestors”

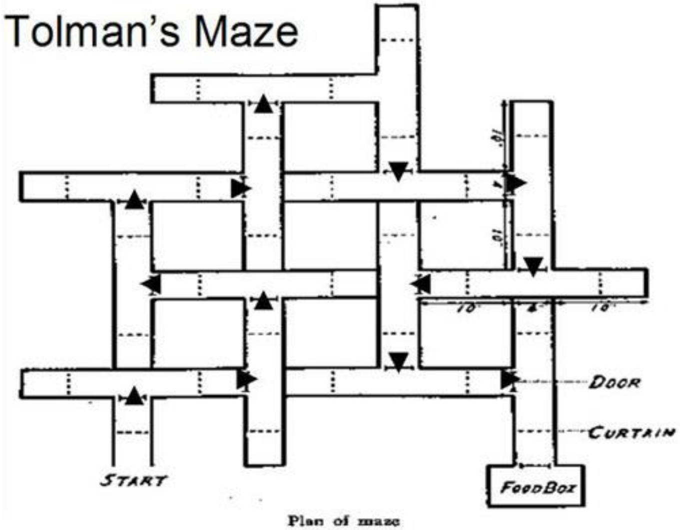
Humans have more complex social lives that does not always correspond with the biological constraints in obvious ways.



Associative learning: Operant conditioning

- BUT ⟶ learning also occurs in the absence of responses.
  - Tolman (1948):
    - Learning involves not only a change in behavior but also the acquisition of new knowledge.
      - *Cognitive map*
      - *Latent learning*





Edward C. Tolman (1886-1959)

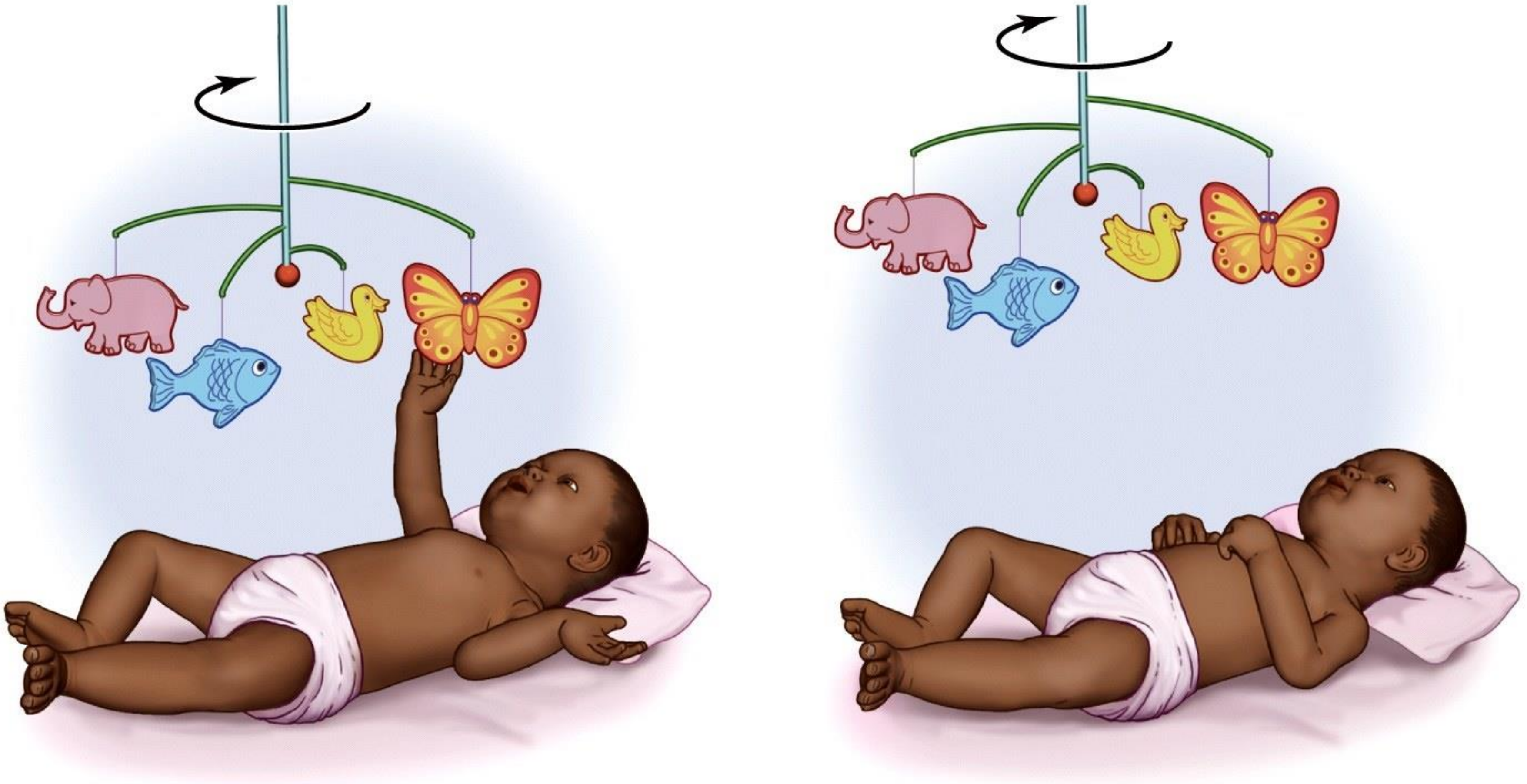
Associative learning: Operant conditioning

- BUT → conditioning requires
  - **Contiguity** (together in time) – learning only occurs when the reinforcer or punishment immediately follows the response.

Associative learning: Operant conditioning

- BUT → conditioning requires
  - **Contiguity** (together in time)
    - learning only occurs when the reinforcer or punishment immediately follows the response.
  - **Contingency**
    - the reinforcer or punishment should occur more likely after the response than otherwise.

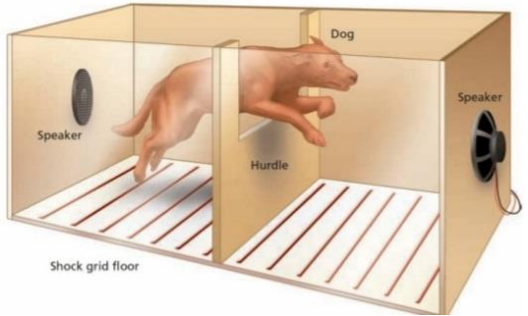
Associative learning: Operant conditioning

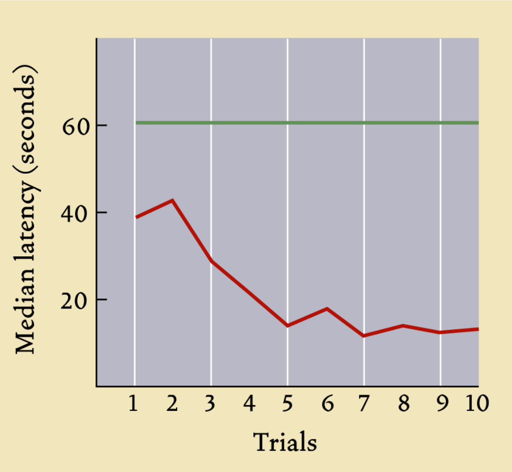


Associative learning: Operant conditioning

- BUT ⟶ conditioning requires
  - Contiguity (together in time)
    - learning only occurs when the reinforcer or punishment immediately follows the response.
  - Contingency
    - the reinforcer or punishment should occur more likely after the response than otherwise
    - Without contingency

      → learned helplessness

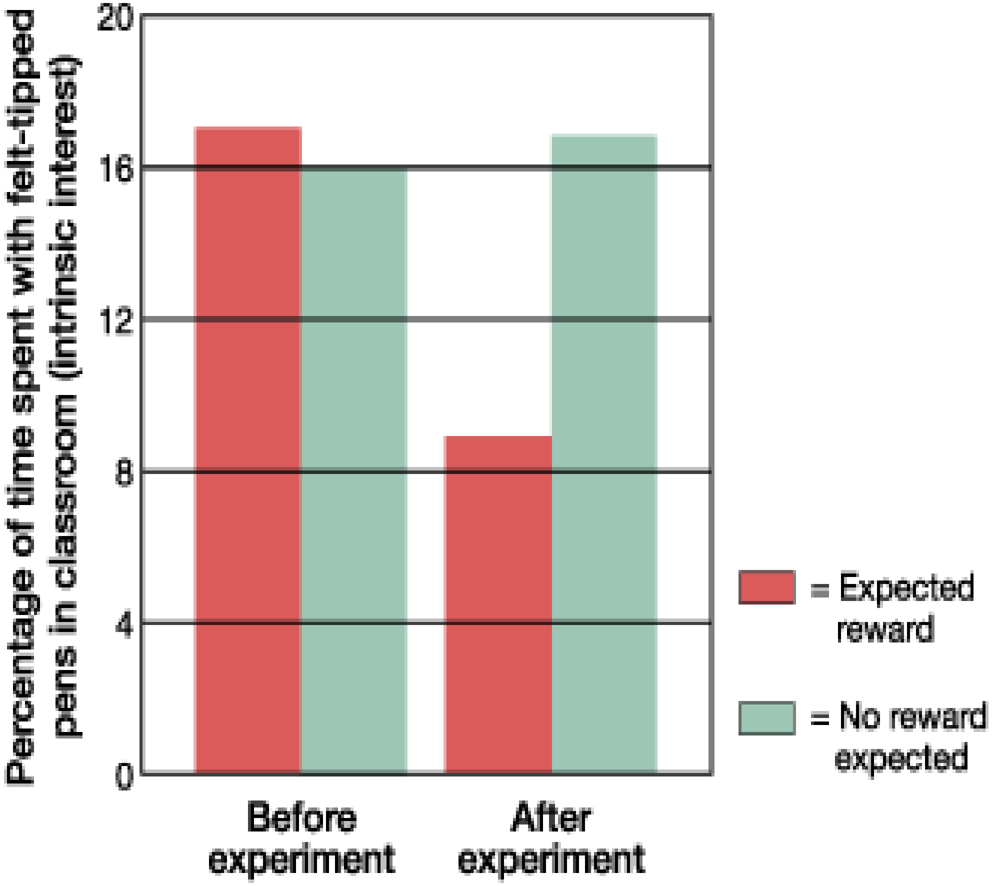




Turning Play Into Work

- When preschoolers were promised a prize for drawing with felt-tip (colorful, bright) pens, the behavior increased.

- After they got the prizes, they spent less time with pens than before the study began.

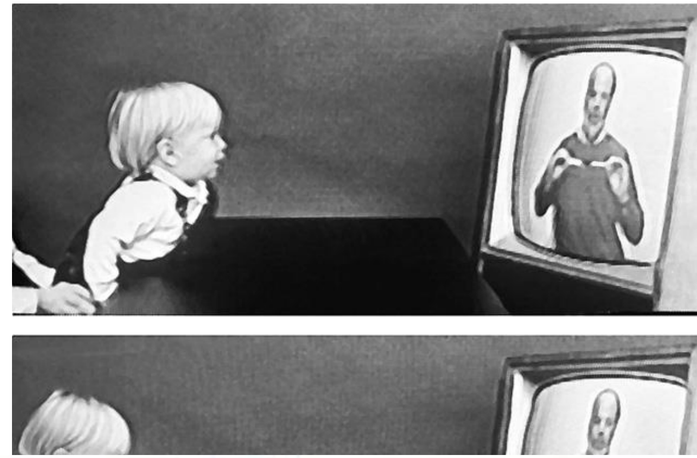


Observational learning

- Observational learning (or social learning):
  - The acquisition or modification of a behavior after exposure to another individual performing that behavior (Bandura's observational studies pg.233)
  - Modeling

Imitation Onset

- Learning by observation begins early in life.
- Shortly after birth, an infant imitates an adult who sticks out his tongue.



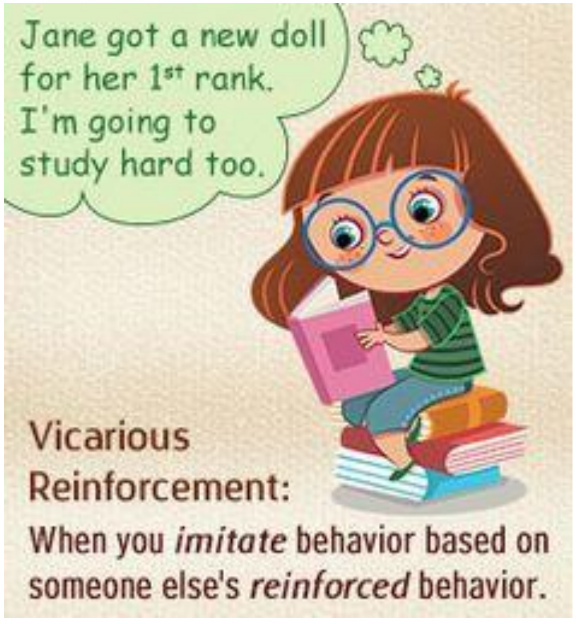
odels by infants.
urtesy of A.N. Meltzoff and M. Hanuk.

Meltzoff, A. N., Kuhl, P. K., Movellan, J. & Sejnowski, T. J. (2009). Foundations for a new science of learning. *Science, 325*, 284–288.

FIGURE 7.15 Imitation This 12-month-old infant sees an adult look left, and immediately follows her gaze (Meltzoff et al., 2009).

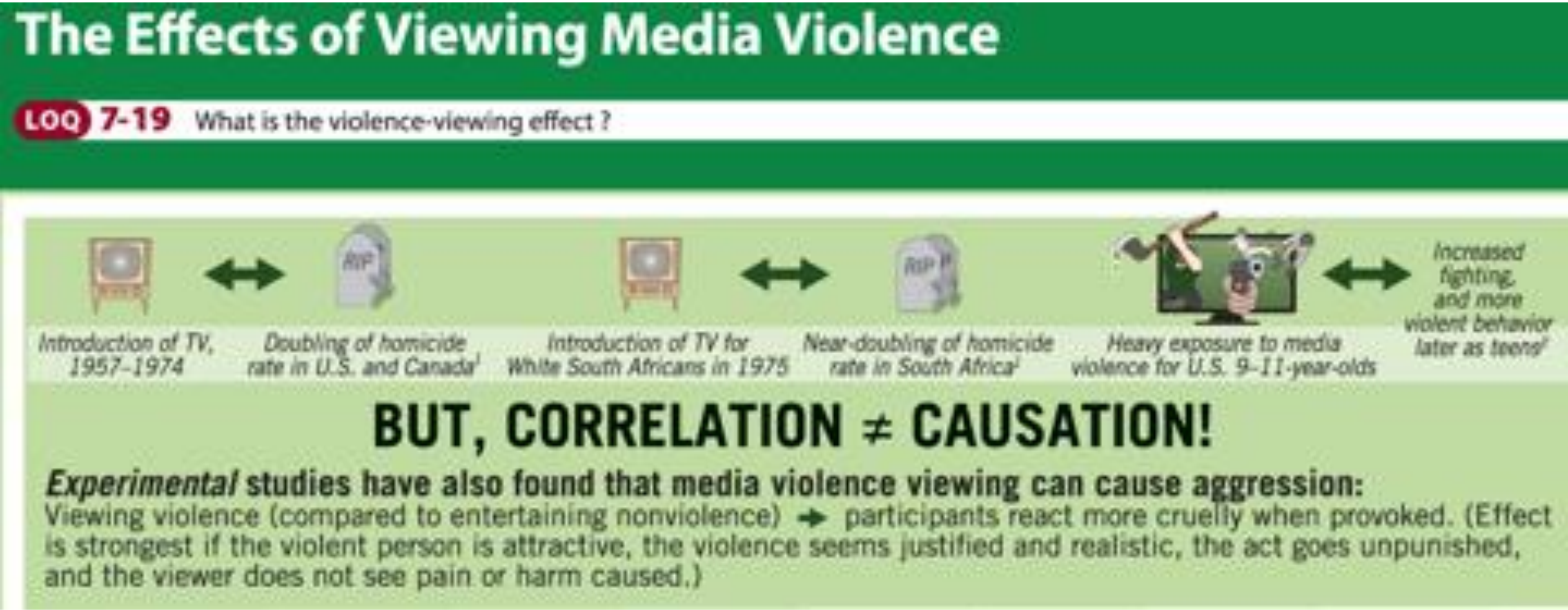
Observational learning

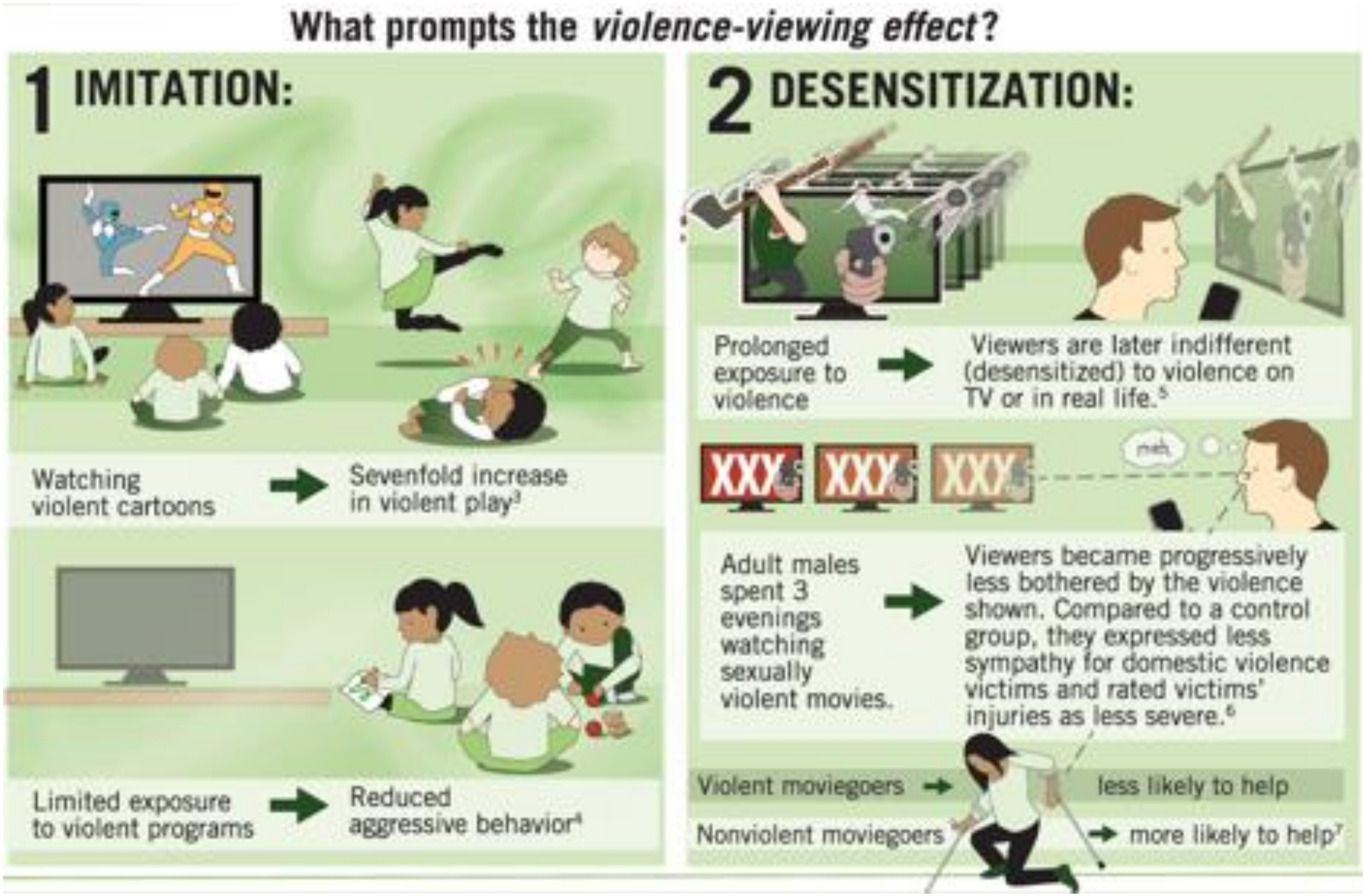
- Observational learning (or social learning):
  - The acquisition or modification of a behavior after exposure to another individual performing that behavior (Bandura's observational studies)
  - Modeling
  - Vicarious learning



Observational learning

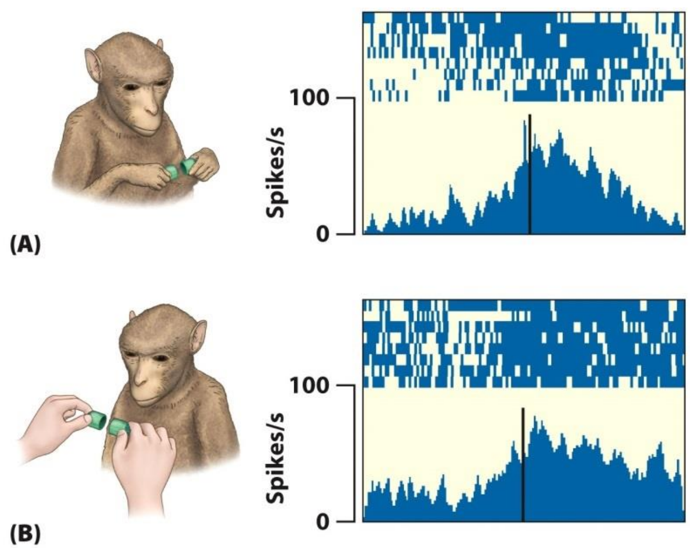
- What about violence in media?




What prompts the *violence-viewing effect*?
1 IMITATION:
Watching violent cartoons
Sevenfold increase in violent play[3]
Limited exposure to violent programs
Reduced aggressive behavior[4]
2 DESENSITIZATION:
Prolonged exposure to violence
Viewers are later indifferent (desensitized) to violence on TV or in real life.[5]
XXX
XXX
XXX
Adult males spent 3 evenings watching sexually violent movies.
Viewers became progressively less bothered by the violence shown. Compared to a control group, they expressed less sympathy for domestic violence victims and rated victims' injuries as less severe.[6]
Violent moviegoers
less likely to help
Nonviolent moviegoers
more likely to help[7]

Observational learning

- What do we know about the mechanisms of observational learning in the brain?
  - Mirror neurons




ME WAITING FOR THIS LECTURE TO END
makeameme.org